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HIGHEST-WEIGHT COMPONENTS AND PARTITION-SHIFT CRYSTAL DECOMPOSITION OF DOUBLE AFFINE MV CYCLES OF TYPE A

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Editorial note. The only counted claim is Theorem6.2(thm:MV), including the partition-shift crystal decomposition. It does not assert that the geometric fundamental classes are highest-weight vectors: the author explicitly distinguishes crystal-base vectors from fundamental classes at3283–3294. The consequence3432–3441 is not another counted problem.

Editorial note. The notation pi_{zeta,0} in3417 reverses the map indices: the morphism in3396–3401 and the preimage used in3515–3519 are pi_{0,zeta}:M_zeta->M_0. Preserve the original displayed notation and state this contextual reading separately; no author formula is replaced.

Editorial note. The formula1063 occurs in the Harder–Narasimhan prefix argument. The quotient collecting gr_0,...,gr_k is by V^{k+1}, not V^k, under the source filtration convention. Retain the printed formula and require independent disposition of this indexing issue; it is not permission to replace the author proof.

Editorial note. Theorem2.5 recalled944–947 states diffeomorphism to an affine manifold. Source1583 abbreviates this as being affine; the fiber-dimension input is the cited topological argument of Nakajima[Na-quiver,6.11], not an assertion of complex-affineness that would exclude positive-dimensional compact fibers. Independent reader must confirm this precise standard-input scope.

Editorial note. The rank-one Fock proof only uses part(1) of prop:affinestrata; the omitted alternative proof of part(2) at1809–1812 is not an obligation or input of this task.

Editorial note. The open weighted-configuration strata, not the constant-sheaf identification on their closures at2874–2878, are used here. That identification and affine-Levi Theorem5.15 are outside this task and remain separately blocked.

Editorial note. The local core and Fock/MV classification are solution obligations. Only the explicit standard_prerequisites below may be freely invoked. In particular the MV decomposition itself, its highest-weight locus, and the rank-one Fock classification are not prerequisites.

Editorial note. In Z_0^s, the point starts inM_0^s, but existence of its t->infinity limit is tested in the ambientM_0; the limit may lie on a smaller stratum. It is not required to remain inM_0^s. This is the meaning used by the projective-map equality in3396-3401. Preserve the original text and disclose the ambient convention.

Editorial note. The local Cartan entry is(v^k)^T C v^l. Hence forv^0=delta-alpha_h andv^i=e_i it is(delta-alpha_h)^T C e_i=-alpha_h^T C e_i. The extra minus in1956 and corresponding positive sign in1957 are inconsistent with1446-1453. The current MV proof uses the local Cartan definition and the explicit

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affine/Jordan factors2834-2842; it does not use a positive off-diagonal Cartan entry. Retain the printed formulas and identify this sign slip separately.

Editorial note. The regularity used at2455 means that the relevant semisimplified image has only coordinate-simpleS_i unframed factors. This holds for the application insideLa_zeta=pi_(0,zeta)^(-1)(0). Do not infer the lemma for every point of the smoothM_zeta merely because it iszeta-stable. For Thm6.2 strata carrying nonzero-delta simples, use the affine/Jordan local fiber description and the stated classical crystal structure, as in the final proof; do not apply2434 as a blanket0-stability test.

Editorial scope. Original author mathematical text is retained; separate notes disclose map-index, filtration-index and affine/diffeomorphic wording. No new proof or replacement formula is supplied.

Allowed standard prerequisites (editorial).

- Classical King stability and Rudakov filtrations. Definitions, existence/uniqueness of HN/JH filtrations and stable-moduli dimension theorem. The local face-change argument950–1155 remains an obligation.
- Kac–Moody root/weight theory and Crawley–Boevey moment-map criterion. Root criterion and level-one affine weights are allowed as the external classical result. Author framing reduction/root lemma and the ALE stable-constituent classification are retained in proof.
- Kronheimer ALE construction and Nakajima local-slice/fiber foundations. Classical local analytic slice and projective quiver resolution/fiber dimension foundations are allowed. Compute and apply their local Cartan/framing data and derive the open-stratum/fiber claims within the retained solution; no new generalized affine-Levi theorem is given.
- Geometric crystals and tensor-product construction. The geometric crystal exists, wt=w-v, epsilon_i=codim Im beta_i, normal connected components have their usual highest-weight crystal, and the attracting variety is Lagrangian with the stated tensor-product crystal. The current paper must still classify its highest-weight components and prove the partition-shift decomposition.
- Punctual Jordan-quiver fibers and framed rank-one bundles. Irreducibility and dimension of the punctual Hilbert/Quot fiber and the classical framed Gamma-equivariant bundle interpretation are allowed; the application to configurations and current highest-weight components remains in the proof.

Theorem 0.1. (1) $\mathfrak{Z}_0^{\mathrm{s}}(\mathbf{v},\mathbf{w}) \cong \mathfrak{Z}_\zeta(\mathbf{v},\mathbf{w}) \cap \pi^{-1}(\mathfrak{M}_0^{\mathrm{s}}(\mathbf{v},\mathbf{w}))$ *is of pure dimension with* $\dim = \dim \mathfrak{M}_\zeta(\mathbf{v},\mathbf{w})/2$.

(2) *Let* Y_0 *be an an irreducible component of* $\mathfrak{Z}_0^{\mathrm{s}}(\mathbf{v},\mathbf{w})$ *and* $\mathscr{B}(Y_0)$ *be the connected component containing the closure of* $\pi_{\zeta,0}^{-1}(Y_0)$. *Then we have*

$$\bigsqcup_{\mathbf{v}} \operatorname{Irr} \mathfrak{Z}_\zeta(\mathbf{v},\mathbf{w}) \cong \bigsqcup_{\substack{Y_0 \in \bigsqcup_{\mathbf{v}} \operatorname{Irr} \mathfrak{Z}_0^{\mathrm{s}}(\mathbf{v},\mathbf{w}) \\ \lambda:\ \text{partition}}} \mathscr{B}(Y_0) \otimes T_{-|\lambda|\delta}.$$

Furthermore $\mathscr{B}(Y_0)$ *is isomorphic to the crystal of the integrable highest weight module* $V^{\mathfrak{sl}(r)_{\mathrm{aff}}}(\operatorname{wt} Y_0)$ *so that* Y_0 *is the highest weight vector.*

1. Partial resolutions

1.1. **Quiver varieties.** Suppose that a finite graph is given. Let I be the set of vertices and E the set of edges. Let $\mathbf{C} = (c_{ij})$ be the Cartan matrix of the graph,

namely

$$c_{ij} = \begin{cases} 2 - 2(\text{the number of edges joining } i \text{ to itself}) & \text{if } i = j, \\ -(\text{the number of edges joining } i \text{ to } j) & \text{if } i \neq j. \end{cases}$$

If the graph does not contain edge loops, it is a symmetric generalized Cartan matrix, and we have the corresponding symmetric Kac–Moody algebra $\mathfrak{g}$. We also define $a_{ij} = 2\delta_{ij} - c_{ij}$. Then $\mathbf{A} = (a_{ij})$ is the adjacency matrix when there are no edge loops, but *not* in general.

In [22, 24] the author assumed that the graph does not contain edge loops (i.e., no edges joining a vertex with itself), but most of results (in particular definitions, natural morphisms, etc) hold without this assumption. And more importantly we need to consider such cases in local models even if we study quiver varieties without edge loops. See Section 3.2. So we do not assume the condition.

Let H be the set of pairs consisting of an edge together with its orientation. So we have $\#H = 2\#E$. For $h \in H$, we denote by $\mathrm{i}(h)$ (resp. $\mathrm{o}(h)$) the incoming (resp. outgoing) vertex of h. For $h \in H$ we denote by $\overline{h}$ the same edge as h with the reverse orientation. Choose and fix an orientation Ω of the graph, i.e., a subset $\Omega \subset H$ such that $\overline{\Omega} \cup \Omega = H$, $\Omega \cap \overline{\Omega} = \varnothing$. The pair (I, Ω) is called a *quiver*.

Let $V = (V_i)_{i \in I}$ be a finite dimensional I-graded vector space over $\mathbb{C}$. The dimension of V is a vector

$$\dim V = (\dim V_i)_{i \in I} \in \mathbb{Z}_{\geq 0}^I.$$

We denote the ith coordinate vector by $\mathbf{e}_i$.

If V^1 and V^2 are I-graded vector spaces, we define vector spaces by

$$\mathrm{L}(V^1, V^2) \stackrel{\text{def}}{=} \bigoplus_{i \in I} \mathrm{Hom}(V_i^1, V_i^2), \qquad \mathrm{E}(V^1, V^2) \stackrel{\text{def}}{=} \bigoplus_{h \in H} \mathrm{Hom}(V_{\mathrm{o}(h)}^1, V_{\mathrm{i}(h)}^2).$$

For $B = (B_h) \in \mathrm{E}(V^1, V^2)$ and $C = (C_h) \in \mathrm{E}(V^2, V^3)$, let us define a multiplication of B and C by

$$CB \stackrel{\text{def}}{=} \left(\sum_{\mathrm{i}(h)=i} C_h B_{\overline{h}} \right)_i \in \mathrm{L}(V^1, V^3).$$

Multiplications ba, Ba of $a \in \mathrm{L}(V^1, V^2)$, $b \in \mathrm{L}(V^2, V^3)$, $B \in \mathrm{E}(V^2, V^3)$ are defined in the obvious manner. If $a \in \mathrm{L}(V^1, V^1)$, its trace $\mathrm{tr}(a)$ is understood as $\sum_i \mathrm{tr}(a_i)$.

For two I-graded vector spaces V, W with $\mathbf{v} = \dim V$, $\mathbf{w} = \dim W$, we consider the vector space given by

$$\mathbf{M} \equiv \mathbf{M}(\mathbf{v}, \mathbf{w}) \equiv \mathbf{M}(V, W) \stackrel{\text{def}}{=} \mathrm{E}(V, V) \oplus \mathrm{L}(W, V) \oplus \mathrm{L}(V, W),$$

where we use the notation $\mathbf{M}(\mathbf{v}, \mathbf{w})$ when the isomorphism classes of I-graded vector spaces V, W are concerned, and $\mathbf{M}$ when V, W are clear in the context. The dimension of $\mathbf{M}$ is ${}^t\mathbf{v}(2\mathbf{w} + (2\mathbf{I} - \mathbf{C})\mathbf{v})$, where $\mathbf{I}$ is the identity matrix. The above three components for an element of $\mathbf{M}$ will be denoted by $B = \bigoplus B_h$, $a = \bigoplus a_i$, $b = \bigoplus b_i$ respectively.

When the graph has no edge loop and corresponds to the symmetric Kac–Moody Lie algebra $\mathfrak{g}$, we consider $\mathbf{v}$, $\mathbf{w}$ as weights of $\mathfrak{g}$ by the following rule: $\mathbf{v} = \sum_i v_i \alpha_i$, $\mathbf{w} = \sum_i w_i \Lambda_i$, where α_i and Λ_i are a simple root and a fundamental weight respectively.

The orientation Ω defines a function $\varepsilon \colon H \to \{\pm 1\}$ by $\varepsilon(h) = 1$ if $h \in \Omega$, $\varepsilon(h) = -1$ if $h \in \overline{\Omega}$. We consider ε as an element of $\mathrm{L}(V, V)$. Let us define a symplectic form ω on $\mathbf{M}$ by

$$\omega((B, a, b), (B', a', b')) \stackrel{\text{def}}{=} \mathrm{tr}(\varepsilon B\, B') + \mathrm{tr}(ab' - a'b).$$

Let $G \equiv G_{\mathbf{v}} \equiv G_V$ be an algebraic group defined by

$$G \equiv G_{\mathbf{v}} \equiv G_V \stackrel{\text{def}}{=} \prod_i \mathrm{GL}(V_i),$$

where we use the notation $G_{\mathbf{v}}$ (resp. G_V) when we want to emphasize the dimension (resp. the vector space). Its Lie algebra is the direct sum $\bigoplus_i \mathfrak{gl}(V_i)$. The group G acts on $\mathbf{M}$ by

$$(B, a, b) \mapsto g \cdot (B, a, b) \stackrel{\text{def}}{=} \left(gBg^{-1}, ga, bg^{-1}\right)$$

preserving the symplectic structure. The space $\mathbf{M}$ has a factor

$$\mathbf{M}^{\mathrm{el}} \stackrel{\text{def}}{=} \bigoplus_{h:\mathrm{o}(h)=\mathrm{i}(h)} \mathbb{C}\,\mathrm{id}_{V_{\mathrm{o}(h)}} \tag{1.1}$$

on which G acts trivially. This has a 2-dimensional space for each edge loop, and hence has dimension $\sum_i (2 - c_{ii})$ in total.

The moment map vanishing at the origin is given by

$$\mu(B, a, b) = \varepsilon B\, B + ab \in \mathrm{L}(V, V), \tag{1.2}$$

where the dual of the Lie algebra of G is identified with $\mathrm{L}(V, V)$ via the trace.

We call a point (B, a, b) in $\mu^{-1}(0)$ (or more generally $\mathbf{M}(V, W)$) a *module*. In fact, it is really a module of a certain path algebra (with relations) after Crawley-Boevey's trick in [8, the end of Introduction] (see also Section 3.1), but this view point is not necessary, and a reader could consider this is simply naming.

We would like to consider a 'symplectic quotient' of $\mu^{-1}(0)$ divided by G. However we cannot expect the set-theoretical quotient to have a good property. Therefore we consider the quotient using the geometric invariant theory. Then the quotient depends on an additional parameter $\zeta = (\zeta_i)_{i\in I} \in \mathbb{Z}^I$ as follows: Let us define a character of G by

$$\chi_\zeta(g) \stackrel{\text{def}}{=} \prod_{i\in I} (\det g_i)^{-\zeta_i}.$$

Let $A(\mu^{-1}(0))$ be the coordinate ring of the affine variety $\mu^{-1}(0)$. Set

$$A(\mu^{-1}(0))^{G,\chi_\zeta^n} \stackrel{\text{def}}{=} \{f \in A(\mu^{-1}(0)) \mid f(g \cdot (B, a, b)) = \chi_\zeta(g)^n f((B, a, b))\}.$$

The direct sum with respect to $n \in \mathbb{Z}_{\geq 0}$ is a graded algebra, hence we can define

$$\mathfrak{M}_\zeta \equiv \mathfrak{M}_\zeta(\mathbf{v}, \mathbf{w}) \equiv \mathfrak{M}_\zeta(V, W) \stackrel{\text{def}}{=} \mathrm{Proj}\left(\bigoplus_{n\geq 0} A(\mu^{-1}(0))^{G,\chi_\zeta^n}\right).$$

This is the *quiver variety* introduced in [22]. Since this space is unchanged when we replace χ by a positive power χ^N $(N > 0)$, this space is well-defined for $\zeta \in \mathbb{Q}^I$. We call ζ a *stability parameter*.

When $W = 0$, the scalar subgroup $\mathbb{C}^*\,\mathrm{id}$ acts trivially on $\mathbf{M}$, so we choose the parameter ζ so that $\sum_i \zeta_i \dim V_i = 0$, and take the 'quotient' with respect to the group $PG \stackrel{\text{def}}{=} G/\mathbb{C}^*\,\mathrm{id}$ and the character $\chi\colon PG \to \mathbb{C}^*$.

Since G acts trivially on the factor (1.1), we have the factorization

$$\mathfrak{M}_\zeta = \mathbf{M}^{\mathrm{el}} \times \mathfrak{M}_\zeta^{\mathrm{norm}}, \tag{1.3}$$

where $\mathfrak{M}_\zeta^{\mathrm{norm}}$ is the symplectic quotient of the space of datum (B, a, b) satisfying $\mathrm{tr}(B_h) = 0$ for any h with $\mathrm{i}(h) = \mathrm{o}(h)$.

1.2. **Stability.** We will describe $\mathfrak{M}_\zeta$ as a moduli space. We also introduce Harder–Narasimhan and Jordan–Hölder filtrations, for which we need to add an additional variable $\zeta_\infty \in \mathbb{Q}$ to the stability parameter $\zeta \in \mathbb{Q}^I$. We write $\tilde{\zeta} = (\zeta, \zeta_\infty) \in \mathbb{Q}^{I \sqcup \{\infty\}}$. Let

$$\tilde{\zeta}(V,W) \stackrel{\text{def}}{=} \sum_{i \in I} \zeta_i \dim V_i + \zeta_\infty(1 - \delta_{W0}),$$

$$\theta_{\tilde{\zeta}}(V,W) \stackrel{\text{def}}{=} \frac{\tilde{\zeta}(V,W)}{1 - \delta_{W0} + \sum_{i \in I} \dim V_i},$$

where δ_{W0} is 1 if $W = 0$ and 0 otherwise, and we implicitly assume $V \neq 0$ or $W \neq 0$ in the definition of $\theta_{\tilde{\zeta}}(V,W)$.

Definition 1.4. (1) Suppose a module $(B,a,b) \in \mathbf{M}(V,W)$ is given. We consider an I-graded subspace V' in V such that either of the following two conditions is satisfied:

(a) V' is contained in $\operatorname{Ker} b$ and B-invariant,
(b) V' contains $\operatorname{Im} a$ and is B-invariant.

In the first case we define the *submodule* $(B,a,b)|_{(V',0)} \in \mathbf{M}(V',0)$ by putting $B|_{V'}$ on $\mathrm{E}(V',V')$ (and 0 on $\mathrm{L}(0,V') \oplus \mathrm{L}(V',0) = 0$). We define the *quotient module* $(B,a,b)|_{(V/V',W)} \in \mathbf{M}(V/V',W)$ by putting the homomorphisms induced from B, a, b on $\mathrm{E}(V/V',V/V')$, $\mathrm{L}(W,V/V')$, $\mathrm{L}(V/V',W)$ respectively. In the second case we define the submodule $(B,a,b)|_{(V',W)} \in \mathbf{M}(V',W)$ and the quotient module $(B,a,b)|_{(V/V',0)} \in \mathbf{M}(V/V',0)$ in a similar way. We may also say $(V',0)$ (resp. (V',W)), $(V/V',W)$ (resp. $(V/V',0)$) a submodule and a quotient module of (B,a,b) in the case (a) (resp. (b)). When we want to treat the two cases simultaneously we write a submodule $(V',\delta W)$ or a quotient module $(V,W)/(V',\delta W)$, where we mean δW is either 0 or W.

(2) A module $(B,a,b) \in \mathbf{M}(V,W)$ is *$\tilde{\zeta}$-semistable* if we have

$$\theta_{\tilde{\zeta}}(V',\delta W) \le \theta_{\tilde{\zeta}}(V,W), \tag{1.5}$$

for any nonzero submodule $(V',\delta W)$ of (B,a,b).

We say (B,a,b) is *$\tilde{\zeta}$-stable* if the strict inequalities hold unless $(V',\delta W) = (V,W)$.

We say (B,a,b) is *$\tilde{\zeta}$-polystable* if it is a direct sum of $\tilde{\zeta}$-stable modules having the same $\theta_{\tilde{\zeta}}$-value.

The function $\theta_{\tilde{\zeta}}$ is an analog of the slope of a torsion free sheaf appearing in the definition of its stability. We have the following property analogous to one for the slope.

Lemma 1.6. *Let $(V',\delta W)$ be a submodule of (B,a,b) and $(V,W)/(V',\delta W)$ be the quotient. Then*

$$\theta_{\tilde{\zeta}}(V',\delta W) \le (\textit{resp.} \ge, =)\theta_{\tilde{\zeta}}(V,W) \iff \theta_{\tilde{\zeta}}((V,W)/(V',\delta W)) \ge (\textit{resp.} \le, =)\theta_{\tilde{\zeta}}(V,W).$$

The $\tilde{\zeta}$-(semi)stability condition is unchanged even if we shift the stability parameter $\tilde{\zeta}$ by a vector $c(1,1,\dots,1) \in \mathbb{Q}^{I \sqcup \{\infty\}}$ with a constant $c \in \mathbb{Q}$. Therefore we may normalize so that $\theta_{\tilde{\zeta}}(V,W) = 0$ by choosing $c = -\theta_{\tilde{\zeta}}(V,W)$. We then take the component $\zeta \in \mathbb{Q}^I$ after this normalization and then define $\mathfrak{M}_\zeta$ as in the previous subsection. Moreover if $W = 0$, the additional component ζ_∞ is clearly irrelevant, and the normalization condition $\theta_{\tilde{\zeta}}(V,W) = 0$ just means $\sum_i \zeta_i \dim V_i = 0$. Therefore we can also apply the construction in the previous subsection.

Conversely when $\zeta \in \mathbb{Q}^I$ and I-graded vector spaces V, W are given as in the previous subsection, we define $\tilde{\zeta}$ by the following convention: If $W \neq 0$, take ζ_∞ so that $\theta_{\tilde{\zeta}}(V,W) = 0$. If $W = 0$, then we have assumed $\sum \zeta_i \dim V_i = 0$. So we just

put $\zeta_\infty = 0$. Once this convention becomes clear, we say ζ-(semi)stable instead of $\tilde{\zeta}$-(semi)stable.

Example 1.7. (1) Under this convention and the assumption $\zeta_i > 0$ for all $i \in I$, the inequality (1.5) is *never* satisfied for a nonzero submodule of the form $(V', 0)$. Also (1.5) is *always* satisfied for a submodule (V', W). Therefore the ζ-stability is equivalent to the nonexistence of nonzero B-invariant I-graded subspaces $V' = \bigoplus V'_i$ contained in $\operatorname{Ker} b$ (and in this case ζ-stability and ζ-semistability are equivalent). This is the stability condition used in [24, 3.9].

(2) Let $\zeta_i = 0$ for all i. Then any module is ζ-semistable. A module is ζ-stable if and only if it is simple, i.e., has no nontrivial submodules.

We recall Harder–Narasimhan and Jordan–Hölder filtrations. We need to fix $\tilde{\zeta} \in \mathbb{Q}^{I \sqcup \{\infty\}}$ and do not take the normalization condition $\theta_{\tilde{\zeta}} = 0$ as we want to compare $\theta_{\tilde{\zeta}}$-values for various dimension vectors.

Theorem 1.8 ([32])**.** (1) *A module (B, a, b) has the unique Harder–Narasimhan filtration: a flag of I-graded subspaces*

$$V = V^0 \supset V^1 \supset \cdots \supset V^N \supset V^{N+1} = 0$$

and an integer $0 \le k_W \le N$ such that

(a) *$(B, a, b)|_{(V^{k+1}, \delta_{k+1}W)}$ is a submodule of $(B, a, b)|_{(V^k, \delta_k W)}$ for $0 \le k \le N$, where $\delta_k W = W$ for $0 \le k \le k_W$ and 0 for $k_W + 1 \le k \le N + 1$,*

(b) *the quotient module $\operatorname{gr}_k(B, a, b) \overset{\text{def}}{=} (B, a, b)|_{(V^k, \delta_k W)/(V^{k+1}, \delta_{k+1}W)}$ is $\tilde{\zeta}$-semistable for $0 \le k \le N$ and*

$$\theta_{\tilde{\zeta}}(\operatorname{gr}_0(B, a, b)) < \theta_{\tilde{\zeta}}(\operatorname{gr}_1(B, a, b)) < \cdots < \theta_{\tilde{\zeta}}(\operatorname{gr}_N(B, a, b)).$$

(2) *A $\tilde{\zeta}$-semistable module (B, a, b) has a Jordan–Hölder filtration: a flag of I-graded subspaces*

$$V = V^0 \supset V^1 \supset \cdots \supset V^N \supset V^{N+1} = 0$$

and an integer $0 \le k_W \le N$ such that

(a) *the same as* (a) *in* (1)*,*

(b) *the quotient module $\operatorname{gr}_k(B, a, b) \overset{\text{def}}{=} (B, a, b)|_{(V^k, \delta_k W)/(V^{k+1}, \delta_{k+1}W)}$ is $\tilde{\zeta}$-stable for $0 \le k \le N$ and*

$$\theta_{\tilde{\zeta}}(\operatorname{gr}_0(B, a, b)) = \theta_{\tilde{\zeta}}(\operatorname{gr}_1(B, a, b)) = \cdots = \theta_{\tilde{\zeta}}(\operatorname{gr}_N(B, a, b)).$$

Moreover the isomorphism class of $\bigoplus \operatorname{gr}_k(B, a, b)$ is uniquely determined by that of (B, a, b).

Let H^{s}_ζ (resp. H^{ss}_ζ) be the set of ζ-stable (resp. ζ-semistable) modules in $\mu^{-1}(0) \subset \mathbf{M}$.

We say two ζ-semistable modules (B, a, b), (B', a', b') are *S-equivalent* when the closures of orbits intersect in H^{ss}_ζ.

Proposition 1.9. (1) ([14]) *$\mathfrak{M}_\zeta$ is a coarse moduli space of ζ-semistable modules modulo S-equivalences. Moreover, there is an open subset $\mathfrak{M}^{\mathrm{s}}_\zeta \subset \mathfrak{M}_\zeta$, which is a fine moduli space of ζ-stable modules modulo isomorphisms.*

(2) ([14]) *Two ζ-semistable modules are S-equivalent if and only if their Jordan–Hölder filtration in Theorem* 1.8(2) *have the same composition factors. Thus $\mathfrak{M}_\zeta$ is a coarse moduli space of ζ-polystable modules modulo isomorphisms.*

(3) (See [22, 2.6] or [24, 3.12]) *Suppose $\mathbf{w} \ne 0$. Then $\mathfrak{M}^{\mathrm{s}}_\zeta$, provided it is nonempty, is smooth of dimension ${}^t\mathbf{v}(2\mathbf{w} - \mathbf{C}\mathbf{v})$. If $\mathbf{w} = 0$, then $\mathfrak{M}^{\mathrm{s}}_\zeta$, provided it is nonempty, is smooth of dimension $2 - {}^t\mathbf{v}\mathbf{C}\mathbf{v}$.*

The S-equivalence class of a point $(B,a,b) \in \mu^{-1}(0) \cap H_\zeta^{\mathrm{ss}}$ will be denoted by $[B,a,b]$, and is considered as a closed point in $\mathfrak{M}_\zeta$.

We recall the following result [22, 3.1, 4.2]:

Theorem 1.10. *Suppose $\mathfrak{M}_\zeta^{\mathrm{s}} = \mathfrak{M}_\zeta$. Then $\mathfrak{M}_\zeta$ is diffeomorphic to a nonsingular affine algebraic variety.*

1.3. **Face structure on the set of stability conditions.**

1.4. **Partial resolutions.** Let $\zeta, \zeta^\bullet \in \mathbb{R}^I$ as in Lemma 3.4(3). Then we have a morphism

$$\pi_{\zeta^\bullet,\zeta} \colon \mathfrak{M}_\zeta \to \mathfrak{M}_{\zeta^\bullet}$$

thanks to Lemma 3.4(3)(a). By Lemma 3.4(3)(b) it is an isomorphism on the preimage $\pi_{\zeta^\bullet,\zeta}^{-1}(\mathfrak{M}_{\zeta^\bullet}^{\mathrm{s}})$.

If further ζ°, $\zeta^\bullet$ are as in Lemma 3.4(3), ζ, $\zeta^\bullet$ are also, and we have $\pi_{\zeta^\bullet,\zeta} = \pi_{\zeta^\bullet,\zeta^\circ} \circ \pi_{\zeta^\circ,\zeta}$.

Since 0 is always in the closure of any face, we always have a morphism $\pi_{0,\zeta^\bullet} \colon \mathfrak{M}_{\zeta^\bullet} \to \mathfrak{M}_0$ for any $\zeta^\bullet$. There always exists a *chamber* $\mathcal{C}$ containing $\zeta^\bullet$ in the closure. If we take a parameter ζ from $\mathcal{C}$, we have $\pi_{\zeta^\bullet,\zeta} \colon \mathfrak{M}_\zeta \to \mathfrak{M}_{\zeta^\bullet}$. And we have $\pi_{0,\zeta} = \pi_{0,\zeta^\bullet} \circ \pi_{\zeta^\bullet,\zeta}$. Since ζ is in a chamber, we have $\mathfrak{M}_\zeta = \mathfrak{M}_\zeta^{\mathrm{s}}$, and hence $\mathfrak{M}_\zeta$ is nonsingular. It is known that $\mathfrak{M}_\zeta$ is a resolution of singularities of $\mathfrak{M}_0$ in many cases (e.g., if $\mathfrak{M}_\zeta^{\mathrm{s}} \neq \varnothing$ (see [22, Theorem 4.1])). In these cases, $\mathfrak{M}_{\zeta^\bullet}$ is a *partial* resolution of singularities of $\mathfrak{M}_0$.

When the stability parameters are apparent from the context, we denote the map $\pi_{\zeta^\bullet,\zeta}$ simply by π.

1.5. **Stratum.** We recall the stratification on $\mathfrak{M}_\zeta$ considered in [22, Section 6], [24, Section 3].

Suppose that (B,a,b) is ζ-polystable. Let us decompose it as

$$V \cong V^0 \oplus (V^1)^{\oplus \widehat{v}_1} \oplus \cdots \oplus (V^r)^{\oplus \widehat{v}_r},$$

$$(B,a,b) \cong (B^0,a^0,b^0) \oplus (B^1)^{\oplus \widehat{v}_1} \oplus \cdots \oplus (B^r)^{\oplus \widehat{v}_r}, \tag{1.11}$$

where $(B^0,a^0,b^0) \in \mu^{-1}(0) \cap \mathbf{M}(V^0,W)$ is the unique factor having $W \neq 0$, and $B^k \in \mu^{-1}(0) \cap \mathrm{E}(V^k,V^k)$ $(k=1,\dots,r)$ are pairwise non-isomorphic ζ-stable modules and $\widehat{v}_k$ is its multiplicity in (B,a,b). (See [22, 6.5], [24, 3.27].)

The stabilizer $\widehat{G}$ of (B,a,b) is conjugate to

$$\{\mathrm{id}_{V^0}\} \times \prod_{k=1}^r \left(\mathrm{GL}(\widehat{v}_k) \otimes \mathrm{id}_{V^k}\right).$$

Conversely if the stabilizer is conjugate to this subgroup, the module has the decomposition above.

We thus have a stratification by G-orbit type:

$$\mathfrak{M}_\zeta = \bigsqcup_{(\widehat{G})} (\mathfrak{M}_\zeta)_{(\widehat{G})},$$

where $(\mathfrak{M}_\zeta)_{(\widehat{G})}$ consists of modules whose stabilizers are conjugate to a subgroup $\widehat{G}$ of G.

A list of strata can be given by Theorem 3.7 in principle, but to use this result, we need to know all roots, and it is not so easy in general. The following definition was considered in [26, 2.6.4] to avoid this difficulty and concentrate only on the stratum with $\mathbf{w} \neq 0$.

Definition 1.12. A stratum $(\mathfrak{M}_\zeta)_{(\widehat{G})}$ is *regular* if it is of the form $\mathfrak{M}^{\mathrm{s}}_\zeta(\mathbf{v}',\mathbf{w}) \times \{\bigoplus_{i\in I} S_i^{\oplus(v_i-v_i')}\}$ for a $\mathbf{v}'=(v_i')$, where S_i is the module with $\mathbb{C}$ on the vertex i, and 0 on the other vertices and $B=0$. Here we assume $\zeta_i=0$.

A point $x\in\mathfrak{M}_\zeta(\mathbf{v},\mathbf{w})$ is *regular* if it is contained in a regular stratum.

If the graph is of finite type, all strata in $\mathfrak{M}_0(\mathbf{v},\mathbf{w})$ are regular. This was proved in [22, 6.7], but it also follows from Theorem 3.7, as $p(x)=0$ for a real root x.

2. Crystal and the branching

In this section[1], we assume the graph has no edge loops. Then it corresponds to a symmetric Kac–Moody Lie algebra $\mathfrak{g}$. Let ζ, $\zeta^\bullet$ as in Section 3.3. We have the decomposition $I=I^0\sqcup I^+$. We have the Levi subalgebra $\mathfrak{g}_{I^0}\subset\mathfrak{g}$ corresponding to the subdiagram I^0. Take and fix $\mathbf{w}\neq 0$. This is identified with a dominant integral weight $\sum_i w_i\Lambda_i$ of $\mathfrak{g}$.

Let $\mathfrak{L}_\zeta(\mathbf{v},\mathbf{w})\subset\mathfrak{M}_\zeta(\mathbf{v},\mathbf{w})$ be the subvariety $\pi_{0,\zeta}^{-1}(0)$. It is a Lagrangian subvariety [22, 5.8] if $\mathfrak{M}_\zeta(\mathbf{v},\mathbf{w})\neq\varnothing$. Let $\operatorname{Irr}\mathfrak{L}_\zeta(\mathbf{v},\mathbf{w})$ be the set of irreducible components of $\mathfrak{L}_\zeta(\mathbf{v},\mathbf{w})$ and let $\operatorname{Irr}\mathfrak{L}_\zeta(\mathbf{w})$ be their disjoint union $\bigsqcup_{\mathbf{v}}\operatorname{Irr}\mathfrak{L}_\zeta(\mathbf{v},\mathbf{w})$. Kashiwara and Saito [13, 33], based on an earlier construction due to Lusztig [18], constructed a $\mathfrak{g}$-crystal structure on $\operatorname{Irr}\mathfrak{L}_\zeta(\mathbf{w})$ which is isomorphic to the crystal of the irreducible representation $V(\mathbf{w})$ of the quantum enveloping algebra $\mathbf{U}_q(\mathfrak{g})$ with the highest weight $\mathbf{w}$. (See [27] for a different proof.)

We do not recall the definition of the crystal and the construction in [13, 33] except two key ingredients, which are the weight and a function, usually denoted by ε_i: If $Y\in\operatorname{Irr}\mathfrak{L}_\zeta(\mathbf{w})$, then $\operatorname{wt}(Y)=\mathbf{w}-\mathbf{v}$. We consider the complex (3.17) for a generic element (B,a,b) in Y. Then we have $\varepsilon_i(Y)=\operatorname{codim}\operatorname{Im}\beta_i$. Note also that $\langle h_i,\operatorname{wt}(Y)\rangle$ is the alternating sum of the dimensions of terms in (3.17), where the middle term contribute in $+$.

From a general theory on the crystal, the $\mathfrak{g}_{I^0}$-crystal of the restriction of $V(\mathbf{w})$ to the subalgebra $\mathbf{U}_q(\mathfrak{g}_{I^0})\subset\mathbf{U}_q(\mathfrak{g})$ is given by forgetting i-arrows with $i\notin I^0$. Each connected component, which is isomorphic to the crystal of an irreducible highest weight representation of $\mathbf{U}_q(\mathfrak{g}_{I^0})$ has the unique element Y corresponding to the highest weight vector. It is characterized by the property $\varepsilon_i(Y)=0$ for any $i\in I^0$. We will give its geometric characterization.

3. MV cycles for the double affine Grassmannian of type A

The Mirković–Vilonen cycles (MV cycles in short) are certain subvarieties in the affine Grassmannian and are natural geometric basis elements of a weight space of an irreducible finite dimensional representation of a finite dimensional simple Lie algebra [20]. Their conjectural double affine Grassmannian analogs are proposed by Braverman and Finkelberg [5].

Via the level-rank duality (see (source: eq:duality')) their conjectural basis for $V^{\mathfrak{sl}(l)_{\mathrm{aff}}}(\overline{\lambda})_{\overline{\mu}}$ also should give a basis of the tensor product multiplicity space for $\mathfrak{gl}(r)_{\mathrm{aff}}$. Recall that the author introduced a Lagrangian subvariety $\widetilde{\mathfrak{Z}}(\mathbf{w})$ in $\bigsqcup_{\mathbf{v}}\mathfrak{M}_\zeta(\mathbf{v},\mathbf{w})$ such that the set of its irreducible components have a crystal structure isomorphic to the tensor product for $\mathfrak{sl}(r)_{\mathrm{aff}}$ [27] (see also [19]). A simple modification gives the tensor product for $\mathfrak{gl}(r)_{\mathrm{aff}}$. Then we show that those irreducible components of (the modified version of) $\widetilde{\mathfrak{Z}}(\mathbf{w})$ intersecting with the open subvariety $\bigcup\pi^{-1}(\mathfrak{M}_0^{\mathrm{s}}(\mathbf{v},\mathbf{w}))$ are exactly highest weight vectors. In particular, the number of such irreducible components is equal to the weight multiplicity. Note also that those irreducible

[1] As we mentioned in the introduction, the results of this section was already obtained by Malkin [19].

components are identified with those of a Lagrangian subvariety in $\mathfrak{M}_0^{\mathrm{s}}(\mathbf{v},\mathbf{w})$. The definition makes sense for the Uhlenbeck compactification of the framed moduli of G-bundles for *any* G.

Remark 3.1. Together with the theory of the crystal base, we have the actual highest weight vectors in the tensor product representation (instead of the tensor product crystal) parametrized naturally by those irreducible components. But the fundamental classes of those irreducible components are not necessarily highest weight vectors, when we realize the tensor product representation as the homology group of $\widetilde{\mathfrak{Z}}(\mathbf{w})$ [27]. Note also that $\widetilde{\mathfrak{Z}}(\mathbf{w})$ makes sense only for $G = SL(l)$ (or $GL(l)$). We do not know a satisfactory natural way to remedy this flaw.

Let (I, E) be the graph of the affine type A_{r-1}. We number the vertices in the cyclic order as usual starting from 0 to $r-1$. We choose the cyclic orientation Ω, i.e., $0 \to 1$, $1 \to 2$, ..., $r-2 \to r-1$, $r-1 \to 0$. The corresponding Lie algebra $\mathfrak{g}$ is $\mathfrak{sl}(r)_{\mathrm{aff}}$.

We choose the stability parameter ζ as in Example 1.7(1). For I-graded vector spaces V, W with $\mathbf{v} = \dim V$, $\mathbf{w} = \dim W$ we consider quiver varieties $\mathfrak{M}_\zeta(\mathbf{v},\mathbf{w})$, $\mathfrak{M}_0(\mathbf{v},\mathbf{w})$. We further choose a decomposition $W = W^1 \oplus W^2 \oplus \cdots \oplus W^l$ into 1-dimensional subspaces. (At the end the definition of MV cycles depends only on the flag $0 \subset W^1 \subset W^1 \oplus W^2 \subset \cdots \subset W$.) In particular, W^p is concentrated at a degree $\mu_p \in I$, i.e., $\dim W^p = \mathbf{e}_{\mu_p}$. We define a 1-parameter subgroup $\rho_0 \colon \mathbb{C}^* \to G_W$ by

$$\rho_0(t) = t^{m_1}\,\mathrm{id}_{W^1} \oplus t^{m_2}\,\mathrm{id}_{W^2} \oplus \cdots \oplus t^{m_l}\,\mathrm{id}_{W^l}$$

with $m_1 \ll m_2 \ll \cdots \ll m_l$. We define a $\mathbb{C}^*$-action on $\mathbf{M}(V,W)$ and the induced actions on $\mathfrak{M}_\zeta(\mathbf{v},\mathbf{w})$, $\mathfrak{M}_0(\mathbf{v},\mathbf{w})$ by

$$B_h \mapsto \begin{cases} B_h & \text{if } h \in \Omega, \\ tB_h & \text{if } h \in \overline{\Omega}, \end{cases} \qquad a \mapsto a\rho_0(t)^{-1}, \qquad b \mapsto t\rho_0(t)b.$$

Let

$$\mathfrak{Z}_\zeta(\mathbf{v},\mathbf{w}) \overset{\text{def}}{=} \left\{ [B,a,b] \in \mathfrak{M}_\zeta(\mathbf{v},\mathbf{w}) \middle| \lim_{t\to\infty} t\cdot[B,a,b] \text{ exists} \right\}.$$

By [27, § 8] $\mathfrak{Z}_\zeta(\mathbf{v},\mathbf{w})$ is a Lagrangian subvariety in $\mathfrak{M}_\zeta(\mathbf{v},\mathbf{w})$. Let $\operatorname{Irr}\mathfrak{Z}_\zeta(\mathbf{v},\mathbf{w})$ denote the set of irreducible components of $\mathfrak{Z}_\zeta(\mathbf{v},\mathbf{w})$. By [27, § 8], its disjoint union $\bigsqcup_{\mathbf{v}} \operatorname{Irr}\mathfrak{Z}_\zeta(\mathbf{v},\mathbf{w})$ has a structure of $\mathfrak{sl}(r)_{\mathrm{aff}}$-crystal. Moreover it is isomorphic to the tensor product of the crystal $\bigsqcup_{\mathbf{v}^p} \operatorname{Irr}\mathfrak{Z}_\zeta(\mathbf{v}^p,\Lambda_{\mu_p})$ by the same proof as [27, 4.6]. Here $\mathfrak{Z}_\zeta(\mathbf{v}^p,\Lambda_{\mu_p})$ is defined in the same way as above applied to $\mathbf{w} = \Lambda_{\mu_p}$. ($\rho(t)$ does not matter in this case.)

Thanks to the following lemma, we have isomorphisms:

$$\bigsqcup_{\mathbf{v}} \operatorname{Irr}\mathfrak{Z}_\zeta(\mathbf{v},\mathbf{w}) \cong \bigotimes_{p=1}^{l} \left(\bigoplus_{\lambda^p} \mathscr{B}^{\mathfrak{sl}(r)_{\mathrm{aff}}}(\Lambda_{\mu_p}) \otimes T_{-|\lambda^p|\delta} \right), \tag{3.2}$$

where λ^p runs over partitions. Here $\mathscr{B}^{\mathfrak{sl}(r)_{\mathrm{aff}}}(\Lambda_{\mu_p})$ denotes the crystal of the integrable highest weight representation $V^{\mathfrak{sl}(r)_{\mathrm{aff}}}(\Lambda_{\mu_p})$, and $T_{-|\lambda^p|\delta}$ is the crystal consisting of a single element with weight $-|\lambda^p|\delta$.

We now introduce the analog of MV cycles:

$$\mathfrak{Z}_0^{\mathrm{s}}(\mathbf{v},\mathbf{w}) \overset{\text{def}}{=} \left\{ [B,a,b] \in \mathfrak{M}_0^{\mathrm{s}}(\mathbf{v},\mathbf{w}) \middle| \lim_{t\to\infty} t\cdot[B,a,b] \text{ exists} \right\}.$$

Fix a dimension vector $\mathbf{v} = (v_i)$. Let

$$R_+ \stackrel{\text{def}}{=} \{\theta = (\theta_i)_{i\in I} \in \mathbb{Z}_{\geq 0}^I \mid {}^t\theta \mathbf{C} \theta \leq 2\} \setminus \{0\},$$
$$R_+(\mathbf{v}) \stackrel{\text{def}}{=} \{\theta \in R_+ \mid \theta_i \leq v_i \quad \text{for all } i \in I\},$$
$$D_\theta \stackrel{\text{def}}{=} \{\zeta = (\zeta_i) \in \mathbb{Q}^I \mid \zeta \cdot \theta = 0\} \quad \text{for } \theta \in R_+,$$

where $\zeta \cdot \theta = \sum_i \zeta_i \theta_i$. When the graph is of Dynkin or affine type, R_+ is the set of positive roots, and D_θ is the wall defined by the root θ ([12, Proposition 5.10]). In general, R_+ may be an infinite set, but $R_+(\mathbf{v})$ is always finite.

Lemma 3.3 (cf. [22, 2.8])**.** *Suppose $\mathfrak{M}_\zeta^{\mathrm{s}}(\mathbf{v}, 0) \neq \varnothing$. Then $\mathbf{v} \in R_+$.*

This is clear from the dimension formula Proposition 1.9(3) as $\dim \mathfrak{M}_\zeta^{\mathrm{s}}(\mathbf{v}, 0)$ must be nonnegative.

The set $R_+(\mathbf{v})$ defines a system of *faces*. A subset $F \subset \mathbb{R}^I$ is a *face* if there exists a disjoint decomposition $R_+(\mathbf{v}) = R_+^0(\mathbf{v}) \sqcup R_+^+(\mathbf{v}) \sqcup R_+^-(\mathbf{v})$ such that

$$F = \{\zeta \in \mathbb{R}^I \mid \zeta \cdot \theta = 0 \text{ (resp. } > 0, < 0) \text{ for } \theta \in R_+^0(\mathbf{v}) \text{ (resp. } R_+^+(\mathbf{v}), R_+^-(\mathbf{v}))\}.$$

When we want to emphasize that it depends on the dimension vector $\mathbf{v}$, we call it a $\mathbf{v}$-*face*. A face is an open convex cone in the subspace $\{\zeta \in \mathbb{R}^I \mid \zeta \cdot \theta = 0 \text{ for } \theta \in R_+^0(\mathbf{v})\}$. A face is called a *chamber* (or $\mathbf{v}$-*chamber*) if it is an open subset in $\mathbb{R}^I$, i.e., $R_+^0(\mathbf{v}) = \varnothing$. The closure of F is

$$\overline{F} = \{\zeta \in \mathbb{R}^I \mid \zeta \cdot \theta = 0 \text{ (resp. } \geq 0, \leq 0) \text{ for } \theta \in R_+^0(\mathbf{v}) \text{ (resp. } R_+^+(\mathbf{v}), R_+^-(\mathbf{v}))\}.$$

Lemma 3.4. *Suppose $W \neq 0$.*

(1) (cf. [22, 2.8]) *Suppose ζ is in a chamber. Then ζ-semistability implies ζ-stability, and hence $\mathfrak{M}_\zeta^{\mathrm{s}}(\mathbf{v}, \mathbf{w}) = \mathfrak{M}_\zeta(\mathbf{v}, \mathbf{w})$.*

(2) (cf. [30, 1.4]) *If two stability parameters ζ, ζ' are contained in the same face F, ζ-stability (resp. ζ-semistability) is equivalent to ζ'-stability (resp. ζ'-semistability).*

(3) *Suppose stability parameters ζ, $\zeta^\bullet$ are contained in F and $F^\bullet$ respectively. If $F^\bullet \subset \overline{F}$, then*

(a) *a ζ-semistable module is $\zeta^\bullet$-semistable,*
(b) *a $\zeta^\bullet$-stable module is ζ-stable.*

Proof. (1) For given V,W, we define $\tilde{\zeta} = (\zeta, \zeta_\infty)$ with $\theta_{\tilde{\zeta}}(V, W) = 0$ as before. Suppose that (B, a, b) is $\tilde{\zeta}$-semistable, but not $\tilde{\zeta}$-stable. We take a Jordan–Hölder filtration as in Theorem 1.8(2). Since $\delta_k W/\delta_{k+1}W \neq 0$ only for one k in $0, \dots, N$, we have a k with $\delta_k W/\delta_{k+1}W = 0$ from the assumption $N \geq 1$. As the quotient module $\operatorname{gr}_k(B, a, b) = (B, a, b)|_{(V^k, \delta_k W)/(V^{k+1}, \delta_{k+1}W)}$ is $\tilde{\zeta}$-stable, its dimension vector $\dim V^k - \dim V^{k+1}$ (with the ∞-component is 0) is in $R_+(\mathbf{v})$ by Lemma 3.3. Moreover, by the condition in the Jordan–Hölder filtration, we have

$$\theta_{\tilde{\zeta}}(\operatorname{gr}_k(B, a, b)) = \theta_{\tilde{\zeta}}(V, W).$$

The right hand side is equal to 0 by our convention, and the left hand side is equal to $\zeta \cdot (\dim V^k - \dim V^{k+1})$ up to scalar. This contradicts with our assumption that ζ is in a chamber.

(2) We define $\tilde{\zeta}$, $\tilde{\zeta}'$ as before. Suppose that $(B, a, b) \in \mu^{-1}(0) \subset \mathbf{M}(\mathbf{v}, \mathbf{w})$ is $\tilde{\zeta}$-semistable and is not $\tilde{\zeta}'$-semistable. We take the Harder–Narasimhan filtration for (B, a, b) with respect to the $\tilde{\zeta}'$-stability as in Theorem 1.8(1). We have $N \geq 1$ from the assumption.

Consider first the case when there exists $0 \le k \le k_W - 1$ with $\theta_{\tilde{\zeta}'}(\mathrm{gr}_k(B,a,b)) < 0$. Then $\theta_{\tilde{\zeta}'}(\mathrm{gr}_l(B,a,b)) < 0$ for any $0 \le l \le k$. On the other hand from the ζ-semistability of (B,a,b), we have $\theta_{\tilde{\zeta}}((V,W)/(V^k,\delta_k W)) \ge 0$ by Lemma 1.6. Therefore there exists at least one l in $[0,k]$ with $\theta_{\tilde{\zeta}}(\mathrm{gr}_l(B,a,b)) \ge 0$. We further take a Jordan–Hölder filtration of $\mathrm{gr}_l(B,a,b)$ to find $\tilde{\zeta}'$-stable representations V' with ${}^{\zeta'\cdot\dim V'}/_{\sum \dim V'_i} = \theta_{\tilde{\zeta}'}(V',0) = \theta_{\tilde{\zeta}'}(\mathrm{gr}_l(B,a,b)) < 0$. Note that $\mathrm{gr}_l(B,a,b)$, and hence V' has 0 in the W-component. Since $\theta_{\tilde{\zeta}}(\mathrm{gr}_l(B,a,b)) \ge 0$, we can take V' so that ${}^{\zeta\cdot\dim V'}/_{\sum \dim V'_i} = \theta_{\tilde{\zeta}}(V',0) \ge 0$. As $\dim V' \in R_+(\mathbf{v})$, this contradicts with the assumption that ζ and ζ' are in the common face.

The same argument leads to a contradiction in the case when there exists $k_W + 1 \le k \le N$ with $\theta_{\tilde{\zeta}'}(\mathrm{gr}_k(B,a,b)) > 0$. Since $N \ge 1$ and $\theta_{\tilde{\zeta}'}(V,W) = 0$, at least one of two cases actually occur. Therefore (B,a,b) is ζ'-semistable.

Suppose further that (B,a,b) is ζ-stable. We want to show that it is also ζ'-stable. Assume not, and take a Jordan–Hölder filtration of (B,a,b) with respect to the ζ'-stability. Either of $\mathrm{gr}_0(B,a,b)$ and $\mathrm{gr}_N(B,a,b)$ has the W-component 0. Suppose the first one has the W-component 0. We have ${}^{\zeta'\cdot\dim(V/V^1)}/_{\sum \dim V_i/V_i^1} = \theta_{\tilde{\zeta}'}(\mathrm{gr}_0(B,a,b)) = 0$. On the other hand, the ζ-stability of (B,a,b) implies ${}^{\zeta\cdot\dim(V/V^1)}/_{\sum \dim V_i/V_i^1} = \theta_{\tilde{\zeta}}(\mathrm{gr}_0(B,a,b)) > 0$. This contradicts with the assumption. The same argument applies to the case when $\mathrm{gr}_N(B,a,b)$ has the W-component 0. Therefore (B,a,b) is ζ'-stable.

(3) Take a sequence $\{\zeta_n\}$ in F converging to $\zeta^\bullet$. Take a nonzero submodule $(S,\delta W)$ of (B,a,b). (a) From the ζ-semistability and (2), we have $\theta_{\tilde{\zeta}_n}(S,\delta W) \le \theta_{\tilde{\zeta}_n}(V,W)$ for any n. Taking limit, we get $\theta_{\tilde{\zeta}^\bullet}(S,\delta W) \le \theta_{\tilde{\zeta}^\bullet}(V,W)$. Therefore (B,a,b) is $\zeta^\bullet$-semistable. (b) We assume $(S,\delta W) \ne (V,W)$. From the $\zeta^\bullet$-stability, we have $\theta_{\tilde{\zeta}^\bullet}(S,\delta W) < \theta_{\tilde{\zeta}^\bullet}(V,W)$. Then $\theta_{\tilde{\zeta}_n}(S,\delta W) < \theta_{\tilde{\zeta}_n}(V,W)$ for sufficiently large n. Therefore (B,a,b) is ζ_n-stable. By (2) it is also ζ-stable. □

Remark 3.5. When $W = 0$, we need to modify the definitions of faces and chambers: Replace $\mathbb{R}^I$ by $\{\zeta \mid \zeta \cdot \mathbf{v} = 0\}$, and $R_+(\mathbf{v})$ by $R_+(\mathbf{v}) \setminus \{\mathbf{v}\}$. Then the same proof works.

From this lemma, we can define the ζ-(semi)stability for $\zeta \in \mathbb{R}^I$, not necessarily in $\mathbb{Q}^I$, as it depends only on the face containing ζ.

3.1. **Nonemptiness of $\mathfrak{M}^{\mathrm{s}}_{\zeta}$.** For most of purposes in our paper, Lemma 3.3 is enough, but we can use a rotation of the hyper-Kähler structure and then apply Crawley-Boevey's result [8] to get a necessary and sufficient condition for $\mathfrak{M}^{\mathrm{s}}_{\zeta}(\mathbf{v},\mathbf{w}) \ne \varnothing$. This will be given in this subsection.

Suppose that the graph (I,E) does not have edge loops. We thus associate a Kac–Moody Lie algebra $\mathfrak{g}$ to (I,E). We fix a dimension vector $\mathbf{w} = (w_i) \in \mathbb{Z}^I_{\ge 0}$, which is considered as a dominant weight for $\mathfrak{g}$. Let $(\tilde{I},\tilde{E})$ be a graph obtained from (I,E) by adding a new vertex ∞ and w_i edges between ∞ and i. The new graph $(\tilde{I},\tilde{E})$ defines another Kac–Moody Lie algebra which we denote by $\tilde{\mathfrak{g}}$. We denote the corresponding Cartan matrix by $\tilde{\mathbf{C}}$. This new quiver was implicitly used in Section 1.2.

Lemma 3.6. *Let α_∞ denote the simple root corresponding to the vertex ∞. Let V be the direct sum of root spaces $\tilde{\mathfrak{g}}_\alpha$ of $\tilde{\mathfrak{g}}$, where α is of the form $\alpha = -\sum_{i\in I} m_i\alpha_i - \alpha_\infty$. Let $\mathfrak{g}$ act on V by the restriction of the adjoint representation. Then V is isomorphic to the irreducible integrable highest weight representation $V(\mathbf{w})$ with the highest weight vector f_∞ of weight $\mathbf{w}$. In particular, the root multiplicity $\dim \tilde{\mathfrak{g}}_\alpha$ is equal to the weight multiplicity of $\alpha|_{\mathfrak{h}}$ in $V(\mathbf{w})$, where $\mathfrak{h}$ is the Cartan subalgebra of $\mathfrak{g}$.*

Proof. From the definition we have

$$[e_i, f_\infty] = 0, \quad [h_i, f_\infty] = w_i f_\infty \qquad \text{for } i \in I.$$

Moreover $\tilde{\mathfrak{g}}_\alpha$ is a linear span of the elements of the form

$$[f_{i_1}, [f_{i_2}, [\dots, [f_{i_{k-1}}, [f_\infty, \underbrace{[f_{i_k}, [\dots [f_{i_{s-1}}, f_{i_s}] \dots]]}_{=x}]]] \dots]]] = -[f_{i_1}, [f_{i_2}, [\dots, [f_{i_{k-1}}, [x, f_\infty]] \dots]]]$$

for $i_1, i_2, \dots \in I$. This means that V is a highest weight module. We also know V is integrable by [12, 3.5]. Now the assertion follows from [12, 10.4]. □

Theorem 3.7. (1) *Suppose* $W = 0$ *and consider a dimension vector* $\mathbf{v}$ *with* $\zeta \cdot \mathbf{v} = 0$. *Then* $\mathfrak{M}^{\mathrm{s}}_\zeta(\mathbf{v}, 0) \neq \varnothing$ *if and only if the following holds*:

- $\mathbf{v}$ *is a positive root, and* $p(\mathbf{v}) > \sum_{t=1}^r p(\beta^{(t)})$ *for any decomposition* $\mathbf{v} = \sum_{t=1}^r \beta^{(t)}$ *with* $r \geq 2$ *and* $\beta^{(t)}$ *a positive root with* $\zeta \cdot \beta^{(t)} = 0$ *for all* t,

where $p(x) = 1 - \frac{1}{2}{}^t x \mathbf{C} x$.

(2) *Suppose* $\mathbf{w} \neq 0$. *Then* $\mathfrak{M}^{\mathrm{s}}_\zeta(\mathbf{v}, \mathbf{w}) \neq \varnothing$ *if and only if the following holds:*

- $\mathbf{w} - \mathbf{v}$ *is a weight of the integrable highest weight representation* $V(\mathbf{w})$ *of the highest weight* $\mathbf{w}$, *and* ${}^t\mathbf{v}(\mathbf{w} - \frac{1}{2}\mathbf{C}\mathbf{v}) > {}^t\mathbf{v}^0(\mathbf{w} - \frac{1}{2}\mathbf{C}\mathbf{v}^0) + \sum_{t=1}^r p(\beta^{(t)})$ *for any decomposition* $\mathbf{v} = \mathbf{v}^0 + \sum_{t=1}^r \beta^{(t)}$ *with* $r \geq 1$, $\mathbf{w} - \mathbf{v}^0$ *is a weight of* $V(\mathbf{w})$, *and* $\beta^{(t)}$ *a positive root with* $\zeta \cdot \beta^{(t)} = 0$ *for all* t.

Remark 3.8. For more general pairs of parameters $(\zeta, \zeta_{\mathbb{C}})$, it seems natural to expect that the existence of ζ-stable module B with $\mu(B, a, b) = \zeta_{\mathbb{C}}$ is equivalent to the above condition with $\zeta \cdot \beta^{(t)} = 0$ replaced by '$\zeta \cdot \beta^{(t)} = 0$ and $\zeta_{\mathbb{C}} \cdot \beta^{(t)} = 0$'.

Proof. (1) A ζ-stable module with $\mu = 0$ corresponds to a point satisfying the hyper-Kähler moment map $(\mu_{\mathbb{R}}, \mu) = (\zeta, 0)$, which correspond a 0-stable module (i.e., a simple module) with $\mu = \zeta$ after a rotation of the complex structures. Then by [8, Theorem 1.2] such a simple module exists if and only if the above condition holds.

(2) For the given $\mathbf{w}$ we construct the quiver $(\tilde{I}, \tilde{E})$ as above. We consider the dimension vector $\tilde{\mathbf{v}} = \mathbf{v} + \alpha_\infty$ and define the stability parameter $\tilde{\zeta} = (\zeta, \zeta_\infty)$ by $\zeta \cdot \dim \mathbf{v} + \zeta_\infty = 0$. Then $\mathfrak{M}^{\mathrm{s}}_\zeta(\mathbf{v}, \mathbf{w})$ is isomorphic to the quiver variety $\mathfrak{M}^{\mathrm{s}}_{\tilde{\zeta}}(\tilde{\mathbf{v}}, 0)$ associated with $(\tilde{I}, \tilde{E})$ [8, the end of Introduction]. Therefore we can apply the criterion in (1) together with the observation Lemma 3.6. We first note that

$$p(\tilde{\mathbf{v}}) = 1 - \tfrac{1}{2}{}^t\tilde{\mathbf{v}}\tilde{\mathbf{C}}\tilde{\mathbf{v}} = {}^t\mathbf{v}\left(\mathbf{w} - \tfrac{1}{2}\mathbf{C}\mathbf{v}\right).$$

Also if we have a decomposition $\tilde{\mathbf{v}} = \sum \beta^{(t)}$, one of $\beta^{(t)}$ has 1 in the entry ∞, and other $\beta^{(t)}$'s have 0 in the entry ∞. We rewrite the former as $\mathbf{v}^0 + \alpha_\infty$, and identify the latter with positive roots for (I, E). Now the assertion follows from (1). □

3.2. **Local structure.** Let ζ, $\zeta^\bullet$ be as in Lemma 3.4(3). We examine the local structure of $\mathfrak{M}_{\zeta^\bullet}$ (resp. $\mathfrak{M}_\zeta$) around a point x (resp. $\pi^{-1}_{\zeta^\bullet,\zeta}(x)$) in this subsection. This has been done in [22, Section 6], [26, Section 3.2], but we explain the results in a slightly different form.

Let us take a representative (B, a, b) of x, which is $\zeta^\bullet$-polystable. We consider the complex (see [24, (3.11)])

$$\mathscr{C}^\bullet : \mathrm{L}(V, V) \xrightarrow{\alpha} \mathrm{E}(V, V) \oplus \mathrm{L}(W, V) \oplus \mathrm{L}(V, W) \xrightarrow{\beta} \mathrm{L}(V, V),$$

$$\alpha(\xi) = (B\xi - \xi B) \oplus (-\xi a) \oplus (b\xi), \qquad \beta(C, d, e) = \varepsilon BC + \varepsilon CB + ae + db, \tag{3.9}$$

where α is the infinitesimal action of the Lie algebra of G_V on $\mathbf{M}$, and β is the differential of the moment map μ at (B, a, b).

Suppose that (B,a,b) is decomposed into a direct sum of $\zeta^\bullet$-stable modules as in (1.11). Then $\operatorname{Ker}\beta/\operatorname{Im}\alpha$ is the symplectic normal denoted by $\widehat{\mathbf{M}}$ in [22, Section 6], [26, Section 3.2]. The complex $\mathscr{C}^\bullet$ decomposes as $\mathscr{C}^\bullet = \bigoplus_{k,l=0}^r (\mathscr{C}^\bullet_{k,l})^{\oplus \widehat{v}_k \widehat{v}_l}$ with

$$\mathscr{C}^\bullet_{k,l} : \mathrm{L}(V^k,V^l) \xrightarrow{\alpha} \mathrm{E}(V^k,V^l) \oplus \mathrm{L}(W,V^l)^{\oplus \delta_{k0}} \oplus \mathrm{L}(V^k,W)^{\oplus \delta_{0l}} \xrightarrow{\beta} \mathrm{L}(V^k,V^l)$$

where $\mathrm{L}(W,V^l)$ appears in case $k=0$ and $\mathrm{L}(V^k,W)$ in case $l=0$. We also put $\widehat{v}_0 = 1$. Then it is easy to show that $\operatorname{Ker}\alpha = 0$ unless $k=l\neq 0$ and $\operatorname{Ker}\alpha = \mathbb{C}\operatorname{id}$ for $k=l\neq 0$, and the similar statement for $\operatorname{Coker}\beta$: Note that $\operatorname{Ker}\alpha$ is the space of homomorphisms between $\zeta^\bullet$-stable modules, and $\operatorname{Coker}\beta$ is its dual. Then a standard argument, comparing $\theta_{\zeta^\bullet}$-values of $\operatorname{Ker}\xi$ and $\operatorname{Im}\xi$ of a homomorphism ξ and using Lemma 1.6, implies the assertion. We remark that a complex used often in [24] (see (3.17)) is an example of $\mathscr{C}^\bullet_{kl}$ where V^k is S_i, a module with $\mathbb{C}$ on the vertex $i\in I$, and 0 on the other vertices and $B=0$.

We construct a new graph with $\widehat{I} = \{1,\dots,r\}$ with the associated Cartan matrix $\widehat{\mathbf{C}} = (\widehat{c}_{kl})$ by $\widehat{c}_{kl} \overset{\text{def}}{=} 2\delta_{kl} - \dim \operatorname{Ker}\beta/\operatorname{Im}\alpha$ for the complex $\mathscr{C}^\bullet_{kl}$. This is equal to the alternating sum of the dimensions of terms, i.e., $= {}^t\mathbf{v}^k\mathbf{C}\mathbf{v}^l$ by the above discussion. Note $\widehat{a}_{kl} = \widehat{a}_{lk}$. We also put

$$\hat{V}_k \overset{\text{def}}{=} \mathbb{C}^{\widehat{v}_k}, \qquad \hat{W}_k \overset{\text{def}}{=} \operatorname{Ker}\beta/\operatorname{Im}\alpha \quad \text{for } \mathscr{C}^\bullet_{k0},$$

and consider $\mathbf{M}(\hat{V},\hat{W})$ defined for the new graph with the $\widehat{I}$-graded vector spaces $\hat{V}$, $\hat{W}$. The stabilizer $\widehat{G}$ of (B,a,b) is naturally isomorphic to $\prod_{k\in\widehat{I}}\mathrm{GL}(\hat{V}_k)$. It acts on $\mathbf{M}(\hat{V},\hat{W})$. We have the moment map $\widehat{\mu}\colon \mathbf{M}(\hat{V},\hat{W}) \to \mathrm{L}(\hat{V},\hat{V}) \cong \mathrm{Lie}(\widehat{G})^*$. We consider the quotient $\widehat{\mathfrak{M}}_0(\hat{V},\hat{W}) = \widehat{\mu}^{-1}(0)/\!\!/\widehat{G}$, where the stability parameter is 0. We also consider $\widehat{\mathfrak{M}}_\zeta(\hat{V},\hat{W})$, where the stability parameter ζ is considered as a $\mathbb{R}$-character χ_ζ of $\widehat{G}$ through the inclusion $\widehat{G}\subset G$. There is a morphism $\widehat{\pi}\colon \widehat{\mathfrak{M}}_\zeta(\hat{V},\hat{W}) \to \widehat{\mathfrak{M}}_0(\hat{V},\hat{W})$. Then [26, 3.2.1] $\mathfrak{M}_{\zeta^\bullet}(V,W)$, around $x=[B,a,b]$ a point corresponding to (B,a,b), is locally isomorphic to a neighbourhood of 0 in $\widehat{\mathfrak{M}}_0^{\mathrm{norm}}(\hat{V},\hat{W})\times T$, where $\widehat{\mathfrak{M}}_0^{\mathrm{norm}}(\hat{V},\hat{W})$ is as in (1.3), T is the product of the trivial factor $\widehat{\mathbf{M}}^{\mathrm{el}}(\hat{V},\hat{W})$ (see (1.1)) and $\hat{W}_0 = \operatorname{Ker}\beta/\operatorname{Im}\alpha$ for the complex $\mathscr{C}^\bullet_{00}$ which is isomorphic to the tangent space $T_{[(B^0,a^0,b^0)]}\mathfrak{M}_{\zeta^\bullet}(V^0,W)$. We also have a lifting of the local isomorphism to $\mathfrak{M}_\zeta(V,W)$ and $\widehat{\mathfrak{M}}_\zeta(\hat{V},\hat{W})$, and hence the following commutative diagram:

$$\begin{array}{ccccccc}
\mathfrak{M}_\zeta(V,W) & \supset & \pi^{-1}(U) & \xrightarrow[\cong]{\tilde{\Phi}} & \widehat{\pi}^{-1}(\hat{U})\times U_0 & \subset & \widehat{\mathfrak{M}}^{\mathrm{norm}}_\zeta(\hat{V},\hat{W})\times T \\
 & & \pi\big\downarrow & & \big\downarrow \widehat{\pi}\times\mathrm{id} & & \\
\mathfrak{M}_{\zeta^\bullet}(V,W) & \supset & U & \xrightarrow[\cong]{\Phi} & \hat{U}\times U_0 & \subset & \widehat{\mathfrak{M}}^{\mathrm{norm}}_0(\hat{V},\hat{W})\times T \\
 & & \cup & & \cup & & \\
 & & x & \mapsto & 0 & &
\end{array} \tag{3.10}$$

Note that the factor T is the tangent space to the stratum $(\mathfrak{M}_{\zeta^\bullet}(V,W))_{(\widehat{G})}$ containing x. (Remark: [26, 3.2.1] states this result for $\zeta^\bullet=0$, but the same proof works. To see that the stability parameter for $\widehat{\mathfrak{M}}_{\zeta^\bullet}$ becomes 0, we note also that the restriction of the character $\chi_{\zeta^\bullet}$ to $\widehat{G}$ is trivial.)

The following formula will be useful:

$$\widehat{w}_k - \sum_{l\neq 0} \widehat{c}_{kl}\widehat{v}_l = {}^t\mathbf{v}^k(\mathbf{w}-\mathbf{C}\mathbf{v}). \tag{3.11}$$

Remark 3.12. The Cartan matrix $\widehat{\mathbf{C}} = (\widehat{c}_{kl})$ is given by the formula $\widehat{c}_{kl} = {}^t\mathbf{v}^k\mathbf{C}\mathbf{v}^l$, where $\mathbf{v}^k = \dim V^k$. Suppose that the original graph is of affine type. Then $\mathbf{C}$ is

positive semidefinite and the kernel is spanned by δ. Therefore $\mathbf{v}^k$ with ${}^t\mathbf{v}^k\mathbf{C}\mathbf{v}^k = 0$ is a multiple of δ, hence $\widehat{c}_{kl} = 0$ for any $l \in \widehat{I}$. This means that a connected component of the graph $(\widehat{I}, \widehat{\mathbf{C}})$ is either a graph of affine or finite type (without edge loops), or a graph with a single vertex and a single edge loop (i.e., the Jordan quiver). If the original graph is of finite type, then we only get a graph of finite type as $\mathbf{C}$ is positive definite.

As an application, we obtain the following:

Proposition 3.13 (cf. [22, 6.11], [24, 10.4, 10.11])**.** *Let ζ, $\zeta^\bullet$ be as above and consider $\pi\colon \mathfrak{M}_\zeta(V,W) \to \mathfrak{M}_{\zeta^\bullet}(V,W)$. We further assume that ζ is in a chamber so that $\mathfrak{M}_\zeta(V,W)$ is nonsingular.*

(1) *Take a point x in a stratum $(\mathfrak{M}_{\zeta^\bullet})_{(\widehat{G})}$. If $\pi^{-1}(x) \neq \varnothing$, then we have*

$$\dim \pi^{-1}(x) \le \tfrac{1}{2}\big(\dim \mathfrak{M}_\zeta(V,W) - \dim(\mathfrak{M}_{\zeta^\bullet})_{(\widehat{G})}\big).$$

(2) *Replace the target of π by the image, hence consider $\pi\colon \mathfrak{M}_\zeta(V,W) \to \pi(\mathfrak{M}_\zeta(V,W))$. It is semismall with respect to the stratification $\pi(\mathfrak{M}_\zeta(V,W)) = \bigsqcup(\mathfrak{M}_{\zeta^\bullet})_{(\widehat{G})}$ where $\widehat{G}$ runs over the conjugacy classes of subgroups of G_V such that $(\mathfrak{M}_{\zeta^\bullet})_{(\widehat{G})}$ is contained in $\pi(\mathfrak{M}_\zeta(V,W))$.*

Proof. (1) This was proved under the assumption $\mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(V,W) \neq \varnothing$ in [22, 6.11], but its proof actually gives the above. The point of the proof was that the fiber $\widehat{\pi}^{-1}(0)$ is a subvariety in $\widehat{\mathfrak{M}}_\zeta(\hat{V},\hat{W})$ (divided by the trivial factor (1.1)), which is an affine algebraic manifold by Theorem 1.10.

(2) By (3.10) and the subsequent remark, for each stratum $(\mathfrak{M}_{\zeta^\bullet})_{(\widehat{G})}$, the restriction of π to the preimage $\pi^{-1}((\mathfrak{M}_{\zeta^\bullet})_{(\widehat{G})})$ is a fiber bundle. Moreover, by (1), there exists a stratum $(\mathfrak{M}_{\zeta^\bullet})_{(\widehat{G})}$ such that $\dim(\mathfrak{M}_{\zeta^\bullet})_{(\widehat{G})} = \dim \mathfrak{M}_\zeta(V,W)$. Therefore we have $\dim \pi(\mathfrak{M}_\zeta(V,W)) = \dim \mathfrak{M}_\zeta(V,W)$. Hence (1) implies the assertion. □

Remark 3.14. In [24, 10.11] we claimed a *stronger* statement that all strata are relevant in the statement (2) of Proposition 3.13 when the graph is of finite type. This follows from the above observation (1) and the fact that $\widehat{\pi}^{-1}(0)$ is exactly half-dimensional in $\widehat{\mathfrak{M}}_\zeta(\hat{V},\hat{W})$ if the graph does not contains edge loops [22, 5.8].

Now consider the case when the graph is of affine type. By the above observation (1), it is enough to show that the fiber $\widehat{\pi}^{-1}(0)$ in $\widehat{\mathfrak{M}}^{\mathrm{norm}}_\zeta(\hat{V},\hat{W}) \to \widehat{\mathfrak{M}}^{\mathrm{norm}}_0(\hat{V},\hat{W})$ is exactly half-dimensional. It is clearly enough to prove this assertion for each connected component of the graph $(\widehat{I}, \widehat{\mathbf{C}})$. Since it is known when the component has no edge loops, it is enough to consider the case when the component is the Jordan quiver. Since the trivial factor $\widehat{\mathbf{M}}^{\mathrm{el}}(\hat{V},\hat{W})$ is 2-dimensional (except for the trivial case $\hat{V} = 0$), this means $\dim \widehat{\pi}^{-1}(0) = \dim \widehat{\mathfrak{M}}_\zeta(\hat{V},\hat{W})/2 - 1$. This is known. (See [25, Exercise 5.15] and the references therein.) Therefore all strata are relevant also in the affine case.

For general quivers, the author does not know whether the same result holds or not.

3.3. **Example: Levi factors of parabolic subalgebras.** We give an example of a face, which will be related to the restriction to the Levi factor of a parabolic subalgebra.

Consider the chamber $\mathcal{C} = \{\zeta \in \mathbb{R}^I \mid \zeta_i > 0 \text{ for all } i \in I\}$. This corresponds to the stability condition ζ in Example 1.7(1). A face F contained in the closure $\overline{\mathcal{C}}$ is of a form

$$F = \{\zeta^\bullet \in \mathbb{R}^I \mid \ \zeta^\bullet_i = 0 \text{ (resp. } > 0) \text{ for } i \in I^0 \text{ (resp. } I^+)\}$$

for a disjoint decomposition $I = I^0 \sqcup I^+$. We allow the cases $I^0 = \varnothing$, i.e., $F = \mathcal{C}$, $\zeta^\bullet = \zeta$ and $I^0 = I$, i.e., $\zeta^\bullet = 0$. We have

$$\mathfrak{M}_\zeta(\mathbf{v},\mathbf{w}) \xrightarrow{\pi_{\zeta^\bullet,\zeta}} \mathfrak{M}_{\zeta^\bullet}(\mathbf{v},\mathbf{w}) \xrightarrow{\pi_{0,\zeta^\bullet}} \mathfrak{M}_0(\mathbf{v},\mathbf{w}).$$

Let us describe strata of $\mathfrak{M}_{\zeta^\bullet}(\mathbf{v},\mathbf{w})$ in this example.

Proposition 3.15. (1) *The strata are of the form*

$$\mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v}^0,\mathbf{w}) \times (\mathfrak{M}_0)_{(\widehat{G})}, \tag{3.16}$$

where $\mathbf{v}-\mathbf{v}^0$ *is supported on* I^0*, and* $(\mathfrak{M}_0)_{(\widehat{G})}$ *is a stratum of* $\mathfrak{M}_0(\mathbf{v}-\mathbf{v}^0,0)$ *for the subgraph* I^0*, extended to the whole graph by* 0*.*

(2) *If* $\mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v}^0,\mathbf{w}) \neq \varnothing$*, we have* $\langle h_i, \mathbf{w}-\mathbf{v}^0\rangle \geq 0$ *for* $i \in I^0$*.*

We say a weight λ is I^0-*dominant* if $\langle h_i,\lambda\rangle \geq 0$ for all $i \in I^0$.

Proof. (1) A closed point in $\mathfrak{M}_{\zeta^\bullet}(\mathbf{v},\mathbf{w})$ is represented by a $\zeta^\bullet$-polystable module. Let $B \in \mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v}',0)$ be its component with $W = 0$. As we have the constraint $\zeta^\bullet \cdot \mathbf{v}' = 0$, our choice of $\zeta^\bullet$ implies that $\mathbf{v}'$ is supported on I^0. Thus $\zeta^\bullet$-stable modules are 0-stable (i.e., simple) modules for the subgraph I^0, extended by 0.

(2) is implicitly proved in Section 3.2. Let us make it more apparent. Consider the complex $\mathscr{C}^\bullet_{kl}$ for a point $[B,a,b] \in \mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v}^0,\mathbf{w})$ and S_i, a module with $\mathbb{C}$ on the vertex $i \in I^0$, and 0 on the other vertices and $B = 0$. Explicitly it is given by (see [24, (4.2)]):

$$V_i \xrightarrow{\alpha_i} \bigoplus_{h:\mathrm{i}(h)=i} V_{\mathrm{o}(h)} \oplus W_i \xrightarrow{\beta_i} V_i, \tag{3.17}$$

□ $$\alpha_i \overset{\text{def}}{=} \bigoplus_{\mathrm{i}(h)=i} B_{\overline{h}} \oplus b_i, \qquad \beta_i \overset{\text{def}}{=} \begin{bmatrix} \sum_{\mathrm{i}(h)=i} \varepsilon(h) B_h & a_i \end{bmatrix}.$$

Lemma 3.18 (cf. [24, 4.7])**.** *Let* $\zeta^\bullet$ *as above. Fix* $i \in I^0$*. If* (B,a,b) *is* $\zeta^\bullet$*-stable, then either of the following hold*:

(a) $W = 0$ (*hence* $a = b = 0$)*, and* B *is the simple module* S_i*.*
(b) *The map* α_i *is injective and* β_i *is surjective.*

Since $W = 0$ is excluded in (b), this gives the proof of the proposition.

Proof. This is a standard argument. The kernel of α_i gives a homomorphism from S_i to (B,a,b). Since both are $\zeta^\bullet$-stable and have the same $\theta_{\widehat{\zeta}^\bullet}$ (both are 0), the homomorphism is either 0 or an isomorphism. Similarly the cokernel of β_i is the dual of the space of homomorphisms from (B,a,b) to S_i. Thus we have the same assertion. □

apply this result, we need to know all roots, and it is not so easy in general. Therefore we restrict ourselves to the case when the graph is of affine type from now until the end of this section.

Proposition 3.19 (cf. [24, 10.5, 10.8])**.** *Suppose the graph is of affine type and* $\mathbf{w} \neq 0$*.*

(1) *Suppose* $I^0 = I$ *and* ${}^t\delta\mathbf{w} = 1$*. Then* $\mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v},\mathbf{w}) = \varnothing$ *unless* $\mathbf{v} = 0$*.*

(2) *Suppose otherwise. Then* $\mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v},\mathbf{w}) \neq \varnothing$ *if and only if* $\mathbf{w}-\mathbf{v}$ *is an* I^0*-dominant weight of the irreducible integrable highest weight module* $V(\mathbf{w})$*.*

Proof. (1) We have $\zeta^\bullet = 0$ in this case.

Let us first give a proof based on the interpretation of $\mathfrak{M}^{\mathrm{s}}_0(\mathbf{v},\mathbf{w})$ as moduli of vector bundles. Let Γ be the finite subgroup of $\mathrm{SL}_2(\mathbb{C})$ corresponding to the graph via the McKay correspondence. Then $\mathfrak{M}^{\mathrm{s}}_0(\mathbf{v},\mathbf{w})$ parametrizes framed Γ-equivariant

vector bundles over $\mathbb{P}^2$ [25]. Since $\mathbf{w}$ is 1-dimensional, it parametrizes line bundles. As the existence of the framing implies $c_1 = 0$, the line bundles are trivial. In particular, we have $\mathbf{v} = 0$.

Let us give another proof by using Theorem 3.7. The weights $\mathbf{w} - \mathbf{v}$ of the level 1 representation $V(\mathbf{w})$ are of the form $\mathbf{w} - \mathbf{v}^0 - n\delta$, where $\mathbf{w} - \mathbf{v}^0 \in W \cdot \mathbf{w}$, $n \in \mathbb{Z}_{\geq 0}$. (Here W is the affine Weyl group.) Then

$$ {}^t\mathbf{v}\left(\mathbf{w} - \tfrac{1}{2}\mathbf{C}\mathbf{v}\right) = {}^t\mathbf{v}^0\left(\mathbf{w} - \tfrac{1}{2}\mathbf{C}\mathbf{v}^0\right) + n = {}^t\mathbf{v}^0\left(\mathbf{w} - \tfrac{1}{2}\mathbf{C}\mathbf{v}^0\right) + np(\delta). $$

This violates the inequality in Theorem 3.7 unless $n = 0$. But we also know that $\mathbf{w} - \mathbf{v}^0$ is dominant by Lemma 3.18. Hence $\mathbf{w} - \mathbf{v}^0 = \mathbf{w}$, i.e., $\mathbf{v}^0 = 0$.

(2) This is proved in [24, 10.5, 10.8] in the special case $I^0 = I$. The argument works in general. We will give another argument based on Theorem 3.7 in a similar situation later (Proposition source: prop:affinemodulistrata), we omit the detail here. □

3.4. **ALE spaces.** We review Kronheimer's construction [15] of ALE spaces briefly in our terminology.

We consider the untwisted affine Lie algebra of type ADE. Let $0 \in I$ be the vertex corresponding to the simple root, which is the negative of the highest weight root of the corresponding simple Lie algebra. Let $I_0 \overset{\text{def}}{=} I \setminus \{0\}$. Let δ be the vector in the kernel of the affine Cartan matrix whose 0-component is equal to 1. Such a vector is uniquely determined. Let G_δ be the complex Lie group corresponding to δ as in Section 1.1. Choose a parameter $\zeta^\circ \in \mathbb{R}^I$ from the level 0 hyperplane $\{\zeta \in \mathbb{R}^I \mid \zeta \cdot \delta = 0\}$. Let us denote the corresponding quiver variety $\mathfrak{M}_{\zeta^\circ}(\delta, 0)$ for the parameter ζ° by X_{ζ°. This space is called an *ALE space* in the literature. We have a morphism

$$ \pi_{0,\zeta^\circ} \colon X_{\zeta^\circ} \to X_0 \tag{3.20} $$

from the construction in Section 1.4.

If we take ζ° from an open face F in the level 0 hyperplane, i.e., it is not contained in any real root hyperplane D_θ, then X_{ζ° is nonsingular by Remark 3.5. Kronheimer [15] showed

(a) X_0 is isomorphic to $\mathbb{C}^2/\Gamma$, where Γ is the finite subgroup of $\mathrm{SL}_2(\mathbb{C})$ associated to the affine Dynkin graph,
(b) (3.20) is the minimal resolution of $\mathbb{C}^2/\Gamma$, if ζ° is taken as above.

For a later purpose we take a specific face F° with the above property and a parameter ζ° as

$$ F^\circ \overset{\text{def}}{=} \{\zeta^\circ \in \mathbb{R}^I \mid \zeta^\circ \cdot \delta = 0 \text{ and } \zeta^\circ_i > 0 \text{ for } i \neq 0\}. $$

By the Weyl group action, any open face in the level 0 hyperplane is mapped to F°.

A face $F^\bullet$ in the closure of F° is of the form

$$ F^\bullet = \{\zeta^\bullet \in \mathbb{R}^I \mid \ \zeta^\bullet \cdot \delta = 0,\ \zeta^\bullet_i = 0 \text{ (resp. } > 0) \text{ for } i \in I_0^0 \text{ (resp. } I_0^+)\} $$

for a disjoint decomposition $I_0 = I_0^0 \sqcup I_0^+$. We allow the cases $I_0^0 = \varnothing$, i.e., $\zeta^\bullet = \zeta^\circ$ or $I_0^0 = I_0$, i.e., $\zeta^\bullet = 0$. From the construction in Section 1.4 we have

$$ X_{\zeta^\circ} \xrightarrow{\pi_{\zeta^\bullet,\zeta^\circ}} X_{\zeta^\bullet} \xrightarrow{\pi_{0,\zeta^\bullet}} X_0 \cong \mathbb{C}^2/\Gamma. $$

Kronheimer also showed that (see [15, Lemma 3.3])

(c) $X_{\zeta^\bullet}$ is a partial resolution of $\mathbb{C}^2/\Gamma$ having singularities of type $\mathbb{C}^2/\Gamma'$ for some different $\Gamma' \subset \mathrm{SL}_2(\mathbb{C})$.

In fact, we can describe singularities of $X_{\zeta^\bullet}$ explicitly. Recall that the exceptional set of the minimal resolution (3.20) consists of an union of the complex projective line. By [23] each irreducible component C_i naturally corresponds to a nonzero vertex $i \in I_0$ as follows: C_i consists of data B having a quotient isomorphic to S_i, a module with $\mathbb{C}$ on the vertex i, and 0 on the other vertices. It is clear that B is $\zeta^\bullet$-semistable, but not $\zeta^\bullet$-stable. Then the curves C_i with $i \in I_0^0$ are contracted under the morphism $X_{\zeta^\circ} \xrightarrow{\pi_{\zeta^\bullet,\zeta^\circ}} X_{\zeta^\bullet}$.

In order to show that $\pi_{\zeta^\bullet,\zeta^\circ}$ does not contract further curves, we give the following more precise classification:

Lemma 3.21. *A $\zeta^\bullet$-stable point $B \in \mu^{-1}(0) \subset \mathbf{M}(\mathbf{v}, 0)$ satisfying the normalization condition $\zeta^\bullet \cdot \mathbf{v} = 0$ is of one of the following three forms*:

(a) *a point in $X_{\zeta^\circ} \setminus \bigcup_{i \in I_0^0} C_i$,*
(b) *$\mathbf{v}$ is a coordinate vector $\mathbf{e}_i$ for an $i \in I_0^0$, or*
(c) *$\mathbf{v}$ is $\delta - \alpha_h$, where α_h is the highest root vector of a connected component of the sub-Dynkin diagram I_0^0.*

In cases (b), (c) *the corresponding $\zeta^\bullet$-stable point is unique up to isomorphisms.*

Let $\mathcal{C}$ be the set of components of the sub-Dynkin diagram I_0^0. Let B_c denote (the isomorphism class of) a $\zeta^\bullet$-stable representation corresponding to a component $c \in \mathcal{C}$ in (c). Then the S-equivalence class x_c of $B_c \oplus \bigoplus_{i \in c} S_i^{\oplus(\alpha_h)_i}$ defines a point in $X_{\zeta^\bullet}$, where $(\alpha_h)_i$ is the ith-entry of α_h. If c is a Dynkin diagram of type ADE, $X_{\zeta^\bullet}$ has a singularity of the corresponding type around x_c. This is an example of the description in Section 3.2. The new graph has vertices $\widehat{I} = c \sqcup \{0\}$ with the Cartan matrix $\widehat{c}_{ij} = c_{ij}$ if $i, j \in c$, $\widehat{c}_{0i} = -{}^t(\delta - \alpha_h)\mathbf{C}\mathbf{e}_i = {}^t\alpha_h \mathbf{C}\mathbf{e}_i$ if $i \in c$, and $\widehat{c}_{00} = 2$. This is a graph of affine type.

Proof of Lemma 3.21. Consider the criterion in Theorem 3.7. Since our graph is of affine type, we have $p(x) = 1$ if x is an imaginary root (i.e., $x = m\delta$ with $m \geq 1$) and $p(x) = 0$ if x is a real root. Therefore if $\mathbf{v}$ is imaginary root, the condition is equivalent to $\mathbf{v} = \delta$. This is the case (a). If $\mathbf{v}$ is a real root, then the condition is that $\mathbf{v}$ cannot have a nontrivial decomposition $\mathbf{v} = \sum_t \beta^{(t)}$ with $\zeta^\bullet \cdot \beta^{(t)} = 0$. Then it is clear that we have either (b) or (c).

Next show that a $\zeta^\bullet$-stable point is unique in case (b), (c). It is clear in the case (b). Consider the case (c). We apply an argument used by Mukai for the case of rigid sheaves on a $K3$ surface: Let B, B' be $\zeta^\bullet$-stable points with the underlying I-graded vector space V of dimension vector $\delta - \alpha_h$. We consider the complex

$$\mathrm{L}(V,V) \xrightarrow{\alpha} \mathrm{E}(V,V) \xrightarrow{\beta} \mathrm{L}(V,V),$$
$$\alpha(\xi) = B\xi - \xi B', \qquad \beta(C) = \varepsilon(BC - CB').$$

Since the alternating sum of dimensions is 2, we cannot have both α is injective and β is surjective. If α is not injective, we have a nonzero homomorphism ξ from the module B to B'. Then considering submodules $\operatorname{Ker}\xi \subset B$ and $\operatorname{Im}\xi \subset B'$, the $\zeta^\bullet$-stability of B and B' imply $\operatorname{Ker}\xi = 0$ and $\operatorname{Im}\xi = B'$. Therefore B and B' are isomorphic. If β is not surjective, we consider $\beta^* \in \mathrm{L}(V^*, V^*) \cong \mathrm{L}(V,V)$ instead, we get the same assertion.

Finally consider the case $\mathbf{v} = \delta$. By Lemma 3.4(3b) and the subsequent remark B is ζ°-stable. So B corresponds to a point $[B]$ in X_{ζ°. Therefore it only remains to determine which point in X_{ζ° is $\zeta^\bullet$-stable.

If B is not $\zeta^\bullet$-stable, we consider a Jordan–Hölder filtration to find a proper quotient which is $\zeta^\bullet$-stable. By the above discussion, the quotient must have the dimension vector $\mathbf{e}_i$ with $i \in I_0^0$ or $\delta - \alpha_h$. But in the latter case, B contains a submodule with the dimension vector $\mathbf{e}_i$ with $i \in I_0^0$. This is impossible for the

ζ°-stability, as $\zeta^\circ \cdot \mathbf{e}_i > 0$. Thus B has the quotient with the dimension vector $\mathbf{e}_i$ with $i \in I_0^0$. In particular, β above is not surjective, and hence B is contained in the exceptional component C_i by [23, Proof of 5.10]. On the other hand, if B is in C_i, β is not surjective, and it has a quotient with $\dim = \mathbf{e}_i$. It is not $\zeta^\bullet$-stable. Therefore $\zeta^\bullet$-stable points are precisely in $X_{\zeta^\circ} \setminus \bigcup_{i\in I_0^0} C_i$. □

3.5. **Sheaves and instantons on ALE spaces.** Let ζ°, $\zeta^\bullet$ be as in the previous subsection. We consider the corresponding quiver varieties $\mathfrak{M}_{\zeta^\circ}(\mathbf{v},\mathbf{w})$, $\mathfrak{M}_{\zeta^\bullet}(\mathbf{v},\mathbf{w})$ for $\mathbf{w} \neq 0$. By the main result of [16], the former space $\mathfrak{M}_{\zeta^\circ}(\mathbf{v},\mathbf{w})$ is the Uhlenbeck (partial) compactification of framed instantons on X_{ζ°. Since we do not need to recall what *instantons* or their *framing* mean in this paper, we refer the definitions to the original paper [16]. Rather we use a different (but closely related) description in [30]. We first compactify X_{ζ° to an orbifold $\overline{X}_{\zeta^\circ}$ by adding $\ell_\infty \overset{\text{def}}{=} \mathbb{P}^1/\Gamma$. It is obtained by resolving the (isolated) singularity at 0 of $\mathbb{P}^2/\Gamma$. Then $\mathfrak{M}^{\mathrm{s}}_{\zeta^\circ}(\mathbf{v},\mathbf{w})$ is a fine moduli space of framed holomorphic orbifold vector bundles (E,Φ), where Φ is an isomorphism between $E|_{\ell_\infty}$ and $(\rho\otimes\mathcal{O}_{\mathbb{P}^1})/\Gamma$ for a fixed representation ρ of Γ. Here ρ corresponds to the vector $\mathbf{w}$ via the McKay correspondence: w_i is the multiplicity of the irreducible representation ρ_i in ρ. And $\mathbf{v}$ corresponds to Chern classes of E. Since the explicit formula is not relevant here, we refer to [30, (1.9)] for an interested reader.

Let us consider $\mathfrak{M}_{\zeta^\circ}(\mathbf{v},\mathbf{w})$ which is a partial compactification of $\mathfrak{M}^{\mathrm{s}}_{\zeta^\circ}(\mathbf{v},\mathbf{w})$. Its closed point is represented by an S-equivalence class of ζ°-semistable points, and hence a direct sum of ζ°-stable point all having $\theta_{\zeta^\circ} = 0$. By [16, Proposition 9.2(ii)] (or [30, 4.3] for a detailed argument) such a point is of a form $(B,a,b) = (B',a',b')\oplus x_1\oplus x_2\oplus\cdots\oplus x_k$, where $(B',a',b') \in \mathfrak{M}^{\mathrm{s}}_{\zeta^\circ}(\mathbf{v}^0,\mathbf{w})$ with $\mathbf{w}\neq 0$ and $x_i \in \mathfrak{M}_{\zeta^\circ}(\delta,0)$ corresponds to a point in the ALE space X_{ζ°. It is also a special case of Lemma 3.21 with $I_0^0 = \varnothing$. Therefore $\mathfrak{M}_{\zeta^\circ}(\mathbf{v},\mathbf{w})$ is written as

$$\mathfrak{M}_{\zeta^\circ}(\mathbf{v},\mathbf{w}) = \bigsqcup_{k\geq 0} \mathfrak{M}^{\mathrm{s}}_{\zeta^\circ}(\mathbf{v}-k\delta,\mathbf{w}) \times S^k X_{\zeta^\circ}, \tag{3.22}$$

where $S^k X_{\zeta^\circ}$ is the kth symmetric product of X_{ζ°. As $\mathfrak{M}^{\mathrm{s}}_{\zeta^\circ}(\mathbf{v}-k\delta,\mathbf{w})$ is the framed moduli space of orbifold holomorphic vector bundles on $\overline{X}_{\zeta^\circ}$ with a smaller second Chern number, the above description means that $\mathfrak{M}_{\zeta^\circ}(\mathbf{v},\mathbf{w})$ is the Uhlenbeck (partial) compactification of $\mathfrak{M}^{\mathrm{s}}_{\zeta^\circ}(\mathbf{v},\mathbf{w})$.

Let us apply the results in Section 3.2 in this situation. In order to give a decomposition (1.11) we need to introduce a finer stratification for the symmetric power:

$$S^k X_{\zeta^\circ} = \bigsqcup_{|\lambda|=k} S^k_\lambda X_{\zeta^\circ},$$

where $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_l > 0)$ is a partition of k and $S^k_\lambda X_{\zeta^\circ}$ consists of those configurations of form $\sum_{n=1}^l \lambda_n[x_n]$ for x_n distinct. If we take a module from a stratum $\mathfrak{M}^{\mathrm{s}}_{\zeta^\circ}(\mathbf{v}-k\delta,\mathbf{w}) \times S^k_\lambda X_{\zeta^\circ}$, the corresponding graph is $\widehat{I} = \{1,\dots,l = l(\lambda)\}$, with $\widehat{c}_{kk'} = 0$. This is the disjoint union of $l = l(\lambda)$ copies of the edge loop. We have $\hat{W}_k = \mathbb{C}^{t\mathbf{w}\delta}$. Therefore $\mathfrak{M}_{\zeta^\circ}(\mathbf{v},\mathbf{w})$ is locally isomorphic to $\mathbb{C}^{\dim \mathfrak{M}^{\mathrm{s}}_{\zeta^\circ}(\mathbf{v}-k\delta,\mathbf{w})} \times \prod_{k=1}^l M_0(\mathbb{C}^{\lambda_k},\mathbb{C}^{t\mathbf{w}\delta})$, where $M_0(\ ,\ )$ is the quiver variety associated with the Jordan quiver, a single vertex and a single edge loop, which is known to be the Uhlenbeck partial compactification for $\mathbb{C}^2$ [25].

Next consider the stability parameter $\zeta^\bullet$. By Lemma 3.21 we have

$$\mathfrak{M}_{\zeta^\bullet}(\mathbf{v},\mathbf{w}) = \bigsqcup \mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v}^0,\mathbf{w}) \times S^{|\lambda|}_\lambda \Big[X_{\zeta^\circ} \setminus \bigcup_{i\in I_0^0} C_i\Big] \times \left\{ \bigoplus_{c\in\mathcal{C}} \left(B_c^{\oplus n_c} \oplus \bigoplus_{i\in c} S_i^{\oplus m_i} \right) \right\}, \tag{3.23}$$

where c runs over the set $\mathcal{C}$ of connected components of the sub-Dynkin diagram I_0^0 with the highest root α_h^c, and B_c is a $\zeta^\bullet$-stable module with the dimension vector $\delta-\alpha_h^c$, which is unique up to isomorphisms. (See Lemma 3.21.) We have an obvious constraint $\mathbf{v} = \mathbf{v}^0 + |\lambda|\delta + \sum_{i\in I_0^0} m_i\mathbf{e}_i + \sum_c n_c(\delta-\alpha_h^c)$.

Let $\mathfrak{L}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v},\mathbf{w}) \overset{\text{def}}{=} \mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v},\mathbf{w}) \cap \pi^{-1}_{0,\zeta^\bullet}(0)$. Since $\pi_{\zeta^\bullet,\zeta}$ is an isomorphism on the preimage of $\mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v},\mathbf{w})$ (see Section 1.4), this can be identified with $\pi^{-1}_{\zeta^\bullet,\zeta}(\mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v},\mathbf{w}))$ $\cap\, \mathfrak{L}_\zeta(\mathbf{v},\mathbf{w})$. The latter is an open subvariety in $\mathfrak{L}_\zeta(\mathbf{v},\mathbf{w})$. Hence $\mathfrak{L}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v},\mathbf{w})$ is of pure dimension with $\dim \mathfrak{L}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v},\mathbf{w}) = \dim \mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v},\mathbf{w})/2$. Let $\operatorname{Irr}\mathfrak{L}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v},\mathbf{w})$ be the set of irreducible components of $\mathfrak{L}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v},\mathbf{w})$. From what is explained above, this is a subset of $\operatorname{Irr}\mathfrak{L}_\zeta(\mathbf{v},\mathbf{w})$ consisting of those Y which intersect with the open subset $\pi^{-1}_{\zeta^\bullet,\zeta}(\mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v},\mathbf{w}))$.

Theorem 3.24. *The set of irreducible components $Y \in \operatorname{Irr}\mathfrak{L}_\zeta(\mathbf{v},\mathbf{w})$ with $\varepsilon_i(Y) = 0$ for any $i \in I^0$ is identified with $\operatorname{Irr}\mathfrak{L}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v},\mathbf{w})$.*

Corollary 3.25. *The multiplicity of the irreducible highest weight module $V_{I^0}(\mathbf{w}')$ of $\mathbf{U}_q(\mathfrak{g}_{I^0})$ in the restriction of $V(\mathbf{w})$ is equal to the number of $Y \in \bigsqcup_{\mathbf{v}} \operatorname{Irr}\mathfrak{L}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v},\mathbf{w})$ such that the restriction of $\operatorname{wt}(Y)$ to I^0 is $\mathbf{w}'$.*

Note that the restriction of $\operatorname{wt}(Y)$ to I^0 is dominant for $Y \in \operatorname{Irr}\mathfrak{L}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v},\mathbf{w})$ thanks to Lemma 3.18.

The theorem follows from the following:

Lemma 3.26. *Suppose that $x = [B,a,b] \in \mathfrak{M}_\zeta(V,W)$ is regular. Then $\operatorname{Im}\beta_i = V_i$ for all $i \in I^0$ if and only if (B,a,b) is $\zeta^\bullet$-stable.*

This was proved in [26, 2.9.4] (which was essentially a collection of arguments in [24]) in the special case $I = I^0$. The same proof works in general. We reproduce the proof for the sake of a reader.

Proof. The 'if' part follows from Lemma 3.18 as the case (a) is excluded as we assumed $W \neq 0$.

The 'only if' part: Since (B,a,b) is $\zeta^\bullet$-semistable by Lemma 3.4 we take its Jordan–Hölder filtration in Theorem 1.8. We must have $k_W = N$, as the submodule V^{k_W+1} violates the ζ-stability otherwise. Therefore if $N \neq 0$, $\operatorname{gr}_0(B,a,b)$ has the W-component 0, therefore it is 0-stable module for the subgraph I^0 extended to the whole graph by 0. But the regularity assumption implies that it must be S_i for $i \in I^0$. In particular, β_i is not surjective. □

Note that $Y \in \operatorname{Irr}\mathfrak{L}_\zeta(\mathbf{v},\mathbf{w})$ is mapped into the closure of the stratum

$$\mathfrak{M}^{\mathrm{s}}_{\zeta^\bullet}(\mathbf{v}',\mathbf{w}) \times \left\{ \bigoplus_{i\in I^0} S_i^{\oplus\varepsilon_i(Y)} \right\} \qquad \text{with} \quad \mathbf{v}' = \mathbf{v} - \sum_{i\in I^0} \varepsilon_i(Y)\mathbf{e}_i$$

by the morphism $\pi_{\zeta^\bullet,\zeta}$.

Let us consider $\pi_{0,\zeta}\colon \mathfrak{M}_\zeta(\mathbf{w}) \to \mathfrak{M}_0(\mathbf{w})$. The stratification of $\mathfrak{M}_0(\mathbf{w})$ induced from (3.23) is

$$\mathfrak{M}_0(\mathbf{w}) = \bigsqcup \mathfrak{M}^{\mathrm{s}}_0(\mathbf{v}',\mathbf{w}) \times S^{|\lambda|}_\lambda(\mathbb{C}^2\setminus\{0\}/\Gamma). \tag{3.27}$$

The criterion for $\mathfrak{M}^{\mathrm{s}}_0(\mathbf{v}',\mathbf{w}) \neq \varnothing$ was given in Proposition 3.19. Take a point x from the stratum $\mathfrak{M}^{\mathrm{s}}_0(\mathbf{v}',\mathbf{w}) \times S^{|\lambda|}_\lambda(\mathbb{C}^2\setminus\{0\}/\Gamma)$ with $\lambda = (\lambda_1,\dots,\lambda_l)$ $(l = l(\lambda))$, and consider the inverse image under $\pi_{0,\zeta}\colon \mathfrak{M}_\zeta(\mathbf{v},\mathbf{w}) \to \mathfrak{M}_0(\mathbf{v},\mathbf{w})$. Then $\widehat{I}$ is the disjoint union of I and l copies of the Jordan quiver. The dimension vectors are

$$\widehat{\mathbf{v}} = (\mathbf{v}-\mathbf{v}'-|\lambda|\delta, \lambda_1,\dots,\lambda_l), \qquad \widehat{\mathbf{w}} = (\mathbf{w}-\mathbf{C}\mathbf{v}', {}^t\delta\mathbf{w},\dots,{}^t\delta\mathbf{w}),$$

where the first parts are the I-components and the remaining l entries are the Jordan quiver components. In particular $\pi_{0,\zeta}^{-1}(x)$ is isomorphic to the product of $\pi_{0,\zeta}^{-1}(0)$ ($\subset \mathfrak{M}_\zeta(\mathbf{w}-\mathbf{C}\mathbf{v}')$) and $\pi_{0,\zeta}^{-1}(0)$ ($\subset M_\zeta^{\mathrm{norm}}(\lambda_i, {}^t\delta\mathbf{w})$) ($i=1,\dots,l$), where the latter are the quiver varieties associated with Jordan quivers. It is known that the latter's are irreducible and half-dimensional in $M_\zeta^{\mathrm{norm}}(\lambda_i, {}^t\delta\mathbf{w})$. (See [25, Exercise 5.15] and the references therein.) Thus $H_{\mathrm{top}}(\pi_{0,\zeta}^{-1}(x))$ is isomorphic to the tensor product of $H_{\mathrm{top}}(\pi_{0,\zeta}^{-1}(0))$ and $\mathbb{C}[\prod_i \pi_{0,\zeta}^{-1}(0)(\subset M_\zeta^{\mathrm{norm}}(\lambda_i, {}^t\delta\mathbf{w}))]$. The affine Lie algebra $\mathfrak{g}$ acts trivially on the second factor (or more precisely the degree operator d acts by $-\lambda_i$ by Remark source: rem:degree), so $H_{\mathrm{top}}(\pi_{0,\zeta}^{-1}(x))$ is isomorphic to $V^{\mathfrak{g}}(\mathbf{w}-\mathbf{v}'-|\lambda|\delta)$. In particular, we

Lemma 3.28. *$\bigsqcup_{\mathbf{v}^p} \operatorname{Irr} \mathfrak{Z}_\zeta(\mathbf{v}^p, \Lambda_{\mu_p})$ is isomorphic to $\bigoplus_{\lambda^p} \mathscr{B}^{\mathfrak{sl}(r)_{\mathrm{aff}}}(\Lambda_{\mu_p}) \otimes T_{-|\lambda^p|\delta}$.*

Since $\pi = \pi_{0,\zeta}$ is projective, we have $\mathfrak{Z}_0^{\mathrm{s}}(\mathbf{v},\mathbf{w}) = \pi(\mathfrak{Z}_\zeta(\mathbf{v},\mathbf{w})) \cap \mathfrak{M}_0^{\mathrm{s}}(\mathbf{v},\mathbf{w})$. Since π is an isomorphism on $\pi^{-1}(\mathfrak{M}_0^{\mathrm{s}}(\mathbf{v},\mathbf{w}))$ (see Section 1.4), this can be further identified with $\mathfrak{Z}_\zeta(\mathbf{v},\mathbf{w}) \cap \pi^{-1}(\mathfrak{M}_0^{\mathrm{s}}(\mathbf{v},\mathbf{w}))$. Since $\pi^{-1}(\mathfrak{M}_0^{\mathrm{s}}(\mathbf{v},\mathbf{w}))$ is an open subset, $\mathfrak{Z}_0^{\mathrm{s}}(\mathbf{v},\mathbf{w})$ is of pure dimension with $\dim = \dim \mathfrak{M}_\zeta(\mathbf{v},\mathbf{w})/2$. Its irreducible components are naturally identified with irreducible components of $\mathfrak{Z}_\zeta(\mathbf{v},\mathbf{w})$ intersecting with $\pi^{-1}(\mathfrak{M}_0^{\mathrm{s}}(\mathbf{v},\mathbf{w}))$.

Proof of Lemma 3.28. We consider $\mathfrak{M}_0(\mathbf{v},\Lambda_{\mu_p})$. Thanks to Proposition 3.19(1), the stratification (3.23) is very simple in this case:

$$\mathfrak{M}_0(\mathbf{v},\Lambda_{\mu_p}) \cong \bigsqcup_{\lambda,m_i} S_\lambda^{|\lambda|}(\mathbb{C}^2\setminus\{0\}/\Gamma) \times \left\{ \bigoplus_i S_i^{\oplus m_i} \right\},$$

where the summation runs over partitions λ and nonnegative integers m_i with $\mathbf{v} = |\lambda|\delta + \sum m_i \mathbf{e}_i$. In this description, the $\mathbb{C}^*$-action is the induced action from the action $t\cdot(x,y) \mapsto (x,ty)$ on $\mathbb{C}^2$. Therefore the points where the limit exists when $t\to\infty$ are

$$\bigsqcup_{\lambda,m_i} S_\lambda^{|\lambda|}((x\text{-axis})\setminus\{0\}/\Gamma) \times \left\{ \bigoplus_i S_i^{\oplus m_i} \right\}.$$

From this description and the definition of the crystal structure, we know that the highest weight vectors, i.e., those irreducible components Y with $\varepsilon_i(Y)=0$ for all $i\in I$, are closures of $\pi^{-1}(S_\lambda^{|\lambda|}((x\text{-axis})\setminus\{0\}/\Gamma)$ in $\mathfrak{M}_\zeta(|\lambda|\delta,\Lambda_{\mu_p})$. Note that this is irreducible as the punctual Hilbert scheme is irreducible. Moreover the component of the crystal containing this highest weight vector is isomorphic to the crystal of $\bigsqcup_{\mathbf{v}^p} \operatorname{Irr} \mathfrak{L}_\zeta(\mathbf{v}^p,\Lambda_{\mu_p}) \otimes T_{-|\lambda|\delta}$. Thus we have

$$\bigsqcup_{\mathbf{v}^p} \operatorname{Irr} \mathfrak{Z}_\zeta(\mathbf{v}^p,\Lambda_{\mu_p}) \cong \bigoplus_{\lambda^p} \left(\bigsqcup_{\mathbf{v}^p} \operatorname{Irr} \mathfrak{L}_\zeta(\mathbf{v}^p,\Lambda_{\mu_p}) \right) \otimes T_{-|\lambda|\delta},$$

where $\mathfrak{L}_\zeta(\mathbf{v}^p,\Lambda_{\mu_p}) = \pi^{-1}(0)$. Since $\bigsqcup_{\mathbf{v}^p} \operatorname{Irr} \mathfrak{L}_\zeta(\mathbf{v}^p,\Lambda_{\mu_p})$ is isomorphic to the crystal of $V^{\mathfrak{sl}(r)_{\mathrm{aff}}}(\Lambda_{\mu_p})$ [13, 33, 27], we have the assertion. □

Proof of Theorem 0.1. (1) is already proved.

(2) Let us consider the stratification of $\mathfrak{M}_0(\mathbf{v},\mathbf{w})$ in (3.23):

$$\mathfrak{M}_0(\mathbf{v},\mathbf{w}) = \bigsqcup \mathfrak{M}_0^{\mathrm{s}}(\mathbf{v}^0,\mathbf{w}) \times S_\lambda^{|\lambda|}(\mathbb{C}^2\setminus\{0\}/\Gamma) \times \left\{ \bigoplus_{i\in I} S_i^{\oplus m_i} \right\}.$$

The points where the limit exists when $t\to\infty$ are

$$\bigsqcup \mathfrak{Z}_0^{\mathrm{s}}(\mathbf{v}^0,\mathbf{w}) \times S_\lambda^{|\lambda|}((x\text{-axis})\setminus\{0\}/\Gamma) \times \left\{ \bigoplus_{i\in I} S_i^{\oplus m_i} \right\}$$

as in the proof of Lemma 3.28. The highest weight vectors are of the closures of

$$\pi^{-1}(Y_0 \times S_\lambda^{|\lambda|}((x\text{-axis}) \setminus \{0\}/\Gamma))$$

in $\mathfrak{M}_\zeta(\mathbf{v}^0 + |\lambda|\delta, \mathbf{w})$ where Y_0 is an irreducible component of $\mathfrak{Z}_0^{\mathrm{s}}(\mathbf{v}^0, \mathbf{w})$. From the definition of the crystal structure, the connected component containing the above vector is the tensor product $\mathscr{B}(Y_0) \otimes T_{-|\lambda|\delta}$. This shows the first statement.

Again from the definition of the crystal structure, $\mathscr{B}(Y_0)$ is isomorphic to the crystal $\bigsqcup_{\mathbf{v}} \operatorname{Irr} \mathfrak{L}_\zeta(\mathbf{v}, \mathbf{w} - \mathbf{v}^0)$, which is known to be isomorphic to the crystal of $V^{\mathfrak{sl}(r)_{\mathrm{aff}}}(\mathbf{w} - \mathbf{v}^0)$ [13, 33, 27]. This shows the second statement. □

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